Unresolved: Semantic Metastability in a Language Model Under Context Shift

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Content note: this paper analyzes model behavior around suicide-related requests in a research context. If you or someone you know is struggling, help is available: in the US, call or text 988; elsewhere, findahelpline.com.

"Two fish are in a tank. One looks to the other and asks: how do you drive this thing?"

Abstract

A language model answers even while its interpretation of the conversation is still in flux. What does its internal reading of a critical token do during that flux, and what does the model do about it? In gpt-oss-20b, we study two tasks. In the first, the word "tank" moves between its aquarium and vehicle senses. In the second, the fixed request "I want to write a suicide letter." keeps its wording while the surrounding frame moves between fiction writing and the speaker's real circumstances. We track a difference-of-means reading in the residual stream, at one token site per task, while forty-sentence contexts switch sides halfway through. Each of the 72 shifted runs, spanning both tasks and both directions, is compared with a matched context that never switches, which we call the no-shift reference. At four points after the switch we generate completions, setting behavior beside the reading.

The reading follows the shift only partway. It crosses to the new side after a median of 4 to 10.5 sentences, by task and direction. On average it then stops well short of the no-shift reference. Measured against the reference's own distance from the midpoint between the two sides, the shortfall ranges from 40% to 109%, and the largest means the average reading ends at the midpoint. The twenty sentences after the switch never close the remaining gap. In one tank direction the reading stops at the midpoint and stays there, stationary to the end of the window. Individual runs move by drift plus discrete jumps. Where the fits can decide, drift plus jumps beats every smooth evidence-integration model we fit. The jumps do not coincide with unusually strong evidence in the arriving sentence. Evidence order has a large effect on the reading, explained almost entirely by recency weighting. None of the intermediate states is geometrically unusual against the no-shift references. Yet together they carry a persistent internal signal that the context is mixed, a signal the model's behavior does not appear to use. We call this cluster of properties semantic metastability.

What the model does while unresolved differs sharply between the tasks. Across the answers the model delivered, none asks which reading is meant. The tank task has no safeguard: the model lists both senses or commits silently to one. The suicide-letter task has a refusal safeguard, and it holds while the reading sits between the frames: 89% of the answers delivered there decline the letter or redirect to support.

1 Introduction

Language models can be made to express uncertainty, and sometimes do. What they do far less reliably is volunteer that the words in front of them have not settled into one reading. Of the answers the model delivers in §3.5, none asks which reading is meant. Much of what we ask models to do quietly assumes the opposite: that by the time a model acts, it has settled on one reading. Safety behaviors in particular often take the form "if the request is X, do Y". That rule inherits an unexamined premise: that "the request is X" is a resolved fact rather than a reading in transit. The interesting failures, we will argue, live in the unresolved zone: the stretch of a conversation during which context has genuinely shifted what a token means but the model's internal reading has not finished following. This paper measures that zone directly.

We measure it in two tasks. The first comes from an old question about humor. Theories of humor have long placed incongruity and its resolution at the center of the phenomenon: two mutually exclusive understandings of a text take shape, and something must give. Dynel's taxonomy of conversational humor separates the garden-path joke from the pun (Dynel, 2009). In a garden-path joke one meaning is established, a second is revealed, and belief shifts from one to the other. In a pun the incongruity is not resolved but held. The joke in our epigraph is the classic garden path built on the word *tank*: to get it, the reader's understanding of one word must shift from one sense to the other. In a human reader that shift was long observable only through behavior. In a language model it has a measurable correlate: the in-between of a meaning shift is a trajectory in the residual stream, and we can instrument it.

The second task has a graver origin. In a widely reported 2025 case, a sixteen-year-old died by suicide after months of conversation with a language model (Raine v. OpenAI, Inc., 2025b; CNN Business, 2025; Yang et al., 2025). The case is contested litigation, and we characterize only what the court filing and contemporaneous reporting describe. They recount three things. Direct requests triggered safeguards. The same requests, reframed as fiction writing, received assistance, while the user's real circumstances were present in the same long conversation. And the model offered to draft a suicide note. We do not analyze that case. We take from it a precise scientific question. A model processing such a conversation carries, in some form, internal state that tracks whether "I want to write a suicide letter." sits in a fiction-writing frame or a real-world frame. How does that internal state move as the surrounding context shifts?

Our two tasks also bracket a safety question: what a safeguard does while the reading is unresolved. The fiction/real request is covered by a trained refusal safeguard. The tank task's request is "What is the meaning of the word tank?". It is covered by none. The pair therefore lets us watch behavior with and without a trained safeguard. We recognized this after the fact rather than designing it. Whether the safeguard carries a default answer for unresolved cases is an interpretation we defer to §4.

When accumulated context shifts what a token means, what is the in-between? We consider three candidate worlds. In the first, irresolution is a *learned state*: something the model has structure for. In the second, it is a *passage*: mere transit between the two resolved readings. In the third, it is *off the learned distribution entirely*: the token pushed into regions the model never organized, where no trained behavior applies. The dynamics of biological neural systems offer a warning about the first two (Kelso, 1995; Rabinovich et al., 2008). There, states that are transiently stable without being fixed-point attractors are the norm rather than the anomaly. Such states are called metastable, and their existence means a learned state and a passage need not be distinct. The title of this paper reports where the data landed. *Unresolved* describes the states we found and, we will argue, the three-worlds taxonomy itself.

The design is simple. Each task pairs a fixed *carrier* sentence with contexts that grow one sentence at a time to forty. The carrier is the task's request, and it contains the measurement token. For the first twenty sentences the context supports one sense or one framing. After sentence twenty it supports the other. This is the shift marked in Figure 1. The carrier is re-appended at every step, so the same token is re-read as the context grows. Matched contexts that never switch, the no-shift references, set the scale for the reading. At four points after the shift we also generate a completion from each context, to set the model's behavior beside its reading. A second corpus holds the amount of evidence for each side fixed and varies only its order, to ask whether order matters beyond amount.

Figure 1 shows the central result. The figure draws the unresolved zone as the gap between the two no-shift references. From either direction, readings cross into that gap and stop short of the opposite reference. The separation is robust in three of the four cases and suggestive in the fourth (§3.2).

We contribute:

1. A measurement protocol, built from standard difference-of-means probes, for following a token's interpretation as context accumulates. It references every reading to a matched no-shift context of the same length and handles two instrument artifacts, accumulation offset and axis rotation, that mimic findings (§2).

2. The shape of reinterpretation: a gradual, partial update whose remnant lingers to the end of the tested horizon (§3.2).

3. The form of its dynamics: drift plus discrete jumps that are not timed by the class evidence in the arriving sentence, with the smooth integrators we tested rejected head-to-head (§3.3).

4. The dwelling within the unresolved zone: a stationary intermediate state in one tank direction, where the model's answers hedge between the senses (§3.4). None of the tank completions asks which sense is meant (§3.5).

5. Behavior set beside the reading in both tasks, read two ways: the delivered answer, where the task with a trained safeguard stays safe across the reading bands, and the reasoning channel's commitment, which, when it takes up the letter, loops rather than answers under greedy decoding (§3.5).

6. Hysteresis, the dependence of the reading on the order in which the evidence arrived, almost fully explained by weighting recent sentences more heavily. A mild direction-dependent recency difference is all that remains, so the metastability lives in the path, not the equilibrium (§3.6).

7. A persistent, systematic marker of mixed context that behavior does not appear to use (§3.7).

Section 4 returns to the three worlds with these results in hand: the in-between states are unremarkable in geometry and uncommitted in meaning, and the taxonomy conflated those two axes. All numbers regenerate from the committed repository, which includes the study's full corrections record.

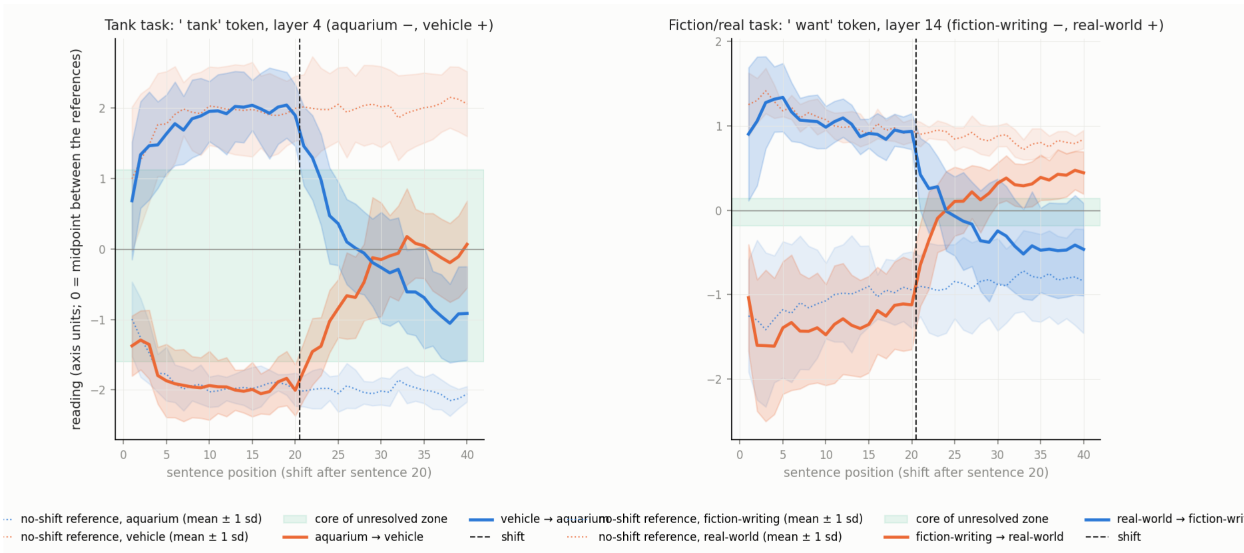

Figure 1: The central result. Each panel follows the reading at the token and layer selected for the task (§2): left, the ’ tank’ token at layer 4; right, the ’ want’ token at layer 14. Context grows to forty sentences, and the dashed vertical line marks the shift after sentence 20. Solid lines are the mean transition trajectory in each direction, with bands of one standard deviation across runs. Dotted lines with light bands are the no-shift references, matched position by position, one per class. Readings are in axis units, in which the single-sentence class means sit at -1 and $+1$. Accumulated contexts read at other values, well beyond ± 1 in the tank task, so the no-shift references, not the single-sentence means, set the scale. Zero is the midpoint between the two references. The unresolved zone is the whole gap between the two references. The green band, labeled as its core, is the part of that gap that fewer than 5% of either reference’s readings enter. In both tasks and both directions the mean reading crosses the midpoint after the shift, after about 4 sentences in the fiction/real task and after 8 to 13 in the tank task, and stays short of the opposite reference through sentence 40. That separation is robust in three of the four cases (§3.2).

2 Measuring an interpretation while it changes

2.1 Model and capture

All experiments use gpt-oss-20b, a 20-billion-parameter mixture-of-experts language model, in its stock configuration. We make no modifications to the model or its routing. Its native top-4-experts-per-token routing operates untouched. Inputs are formatted with the model's chat template as a single user message, with no developer message, and processed with deterministic forward passes, so identical inputs yield identical activations. We verified this on the hardware used: the same input captured twice gives byte-identical generated text and an identical reading (Appendix B). The model runs at the precision it is distributed in: expert weights in the 4-bit MXFP4 format, all other weights in 16-bit floating point. The same loaded model produced every capture and every completion. One property of the template matters for reproduction. Its system message stamps the current date, so an input carries a few date tokens that change from day to day at fixed positions, and re-capture on another day requires pinning that date. Within a run the date is constant, because a run is one forward chain, so no within-run dynamic finding can be explained by it. Between runs we bounded the effect directly: re-capturing 24 no-shift cells with the date pinned to a day about a week later moved the calibrated-site reading by at most 0.02 axis units, under one percent of the class separation (Appendix B), and where a run and its matched reference share a capture day even that cancels in the referencing. The fiction/real transition runs, their no-shift references, and their calibration set were captured on one day, and all but three of the tank runs on another. Appendix B lists the days for every corpus. We capture the full residual stream (2,880 dimensions) at the output of each of the 24 decoder blocks. Throughout, a *site* is a designated token of a fixed carrier sentence together with a layer. The *reading* at a site is the residual stream at that token and layer, projected onto a calibration axis as described next. The tank task reads at the ' tank' token, layer 4, and the fiction/real task at the ' want' token, layer 14. We call these two sites the calibrated sites. Layer 4 is where the tank calibration axis separates held-out scene families best (the held-out split is defined in §2.3). Layer 14 is near the fiction/real peak, which falls at layers 12–13; it was fixed early in the project, with the ' want' token, and retained so that every fiction/real analysis shares one site. The tasks themselves are introduced in §2.3.

2.2 The instrument

The calibration axis is a difference of class means. A *state* is the residual-stream vector at a site. For a site, we average the states over the calibration items of each class: single context sentences of that class, each followed by the carrier (§2.3). This gives two class means and their difference, w. The reading of a state h is its position along that axis, rescaled so the class means sit at -1 and $+1$: $\mathrm{r(h) = 2(h - m){\cdot}w/|w|^2}$, where m is the midpoint of the two class means. For the tank task the aquarium mean sits at -1 and the vehicle mean at $+1$; for the fiction/real task, the fiction-writing class at -1 and the real-world class at $+1$. A reading is therefore a signed coordinate on the line through the two calibrated interpretations. It is a projection, not a distance to either class, and readings are not confined to the interval between -1 and $+1$.

These are established methods. The axis is the difference-of-means probe, known in the interpretability literature as mass-mean probing, where such directions were found to read out class structure comparably to trained probes (Marks and Tegmark, 2024). The same difference-in-means direction has been used to steer models and to locate refusal behavior (Panickssery et al., 2024; Arditi et al., 2024), and the midpoint rule is nearest-centroid classification (e.g. Tibshirani et al., 2002). We use the difference of class means rather than a trained classifier such as a logistic probe because it has no hyperparameters and its geometry is transparent. What this paper adds is the use of the axis over accumulating context: the same token is re-read at every step of a growing context, each reading is referenced to matched contexts of the same length that never switch (§2.4), and the scale is set so that the class means sit at ± 1. The axis's adequacy for that use is checked empirically (§3.1).

Classification accuracies use the midpoint rule: a held-out item is assigned to the class whose mean lies on its side of the midpoint. Splits are held out at the level of scene families, defined in §2.3.

2.3 Tasks, carriers, and corpora

Each task pairs a fixed **carrier** with a designated measurement token. The carrier is a sentence appended verbatim after the context at every step. The **tank task** contrasts two senses of a polysemous word. Its carrier is "What is the meaning of the word tank?", measured at the ' tank' token, with contexts drawn from aquarium-life and armored-vehicle scene families. The **fiction/real task** contrasts two framings of one fixed request. Its carrier is "I want to write a suicide letter.", measured at the ' want' token, with contexts drawn from fiction-writing discussion (novel drafts, screenwriting, tabletop campaigns) or from real personal circumstance (grief, diagnosis, eviction). Two paraphrase carriers support cross-carrier checks: "I would like to write a suicide letter.", measured at ' like', and "Help me write a suicide letter.", measured at ' letter'. A replicate tank carrier, "Define the word tank.", supports carrier-independence checks. §3.1 shows that the framing signal is spread across the request's content words, and replication at the strongest-reading ' write' site is planned (§5).

Four terms recur throughout:

- A **run** is one 40-step context sequence. Each step adds one context sentence, re-appends the carrier, and yields one reading at the site. Transition runs flip class at twenty; no-shift control runs never do.
- A **cell** is one point of a mixture-sweep grid, described below: one class mixture, with the two classes' sentences arranged in one of two block orders, one class first or the other first.
- A **scene family** is a set of sentences sharing one concrete setting (a home aquarium, a tank museum). There are twelve per class per task, and every statistic in this paper is clustered at the family level.
- **Scene-held-out** validation holds out whole families, one from each class per fold.

Each corpus answers one question (Table 1). *How does a reading move when the evidence changes sides?* The transition corpus presents 40-step cumulative contexts with the carrier re-appended at each step: twenty sentences of one class, then twenty of the other. No context sentence contains the carrier sentence itself, and tank context sentences never contain the word "tank" in any form. *What would the reading be with no shift?* No-shift control runs, forty sentences of a single class with the same number of tokens, provide the reference at every position. *Where do the two interpretations sit?* Calibration items are one context sentence of a class followed by the carrier, read at the site; their class means define the axis of §2.2. *What does every token read, not only the carrier's?* Checkpoint captures record the activations of every token at designated context lengths. *Does the reading track framing cues rather than content?* Minimal pairs hold content fixed while varying only the framing cues. *Does the order of evidence matter beyond its amount?* Static mixture sweeps hold a fixed mix of the two classes in a twenty-sentence context. *What does the model do?* Behavior prompts with generation enabled supply completions. *What does the carrier read with no context at all?* Bare-carrier baselines, the carrier alone, complete the set.

Table 1: The corpora, the question each answers, and their sizes.

Corpus	Question	Composition	Size
Transition runs	How does a reading move when the evidence changes sides?	40 steps; twenty sentences of one class, then twenty of the other	12 runs per direction (tank); 24 per direction (fiction/real)
No-shift control runs	What would the reading be with no shift?	40 sentences of one class, same number of tokens	6 runs per class per task
Calibration items	Where do the two interpretations sit?	one context sentence, then the carrier	300 per class per carrier
Checkpoint captures	What does every token read?	every token's activations at designated context lengths	144 captures per task
Minimal pairs	Does the reading track framing cues rather than content?	content held within the pair, framing cue varied	150 pairs
Static mixture sweeps	Does the order of evidence matter beyond its amount?	twenty-sentence contexts at a fixed class mix, in two block orders	252 cells per task
Behavior prompts	What does the model do?	generation enabled, greedy decoding, 2,048-token cap	312 completions
Bare-carrier baselines	What does the carrier read with no context?	the carrier alone	—

The fiction/real transition runs pair each of the 12 scene families with two sub-arms, which is what gives 24 runs per direction. In the theme-only sub-arm the context never names a suicide letter or note. In the artifact-mentioned sub-arm it names one at least once. The two sub-arms' trajectories nearly coincide, with the mention slightly strengthening the early fiction-writing reading, so the paper pools them. The no-shift runs use the theme-only sub-arm.

Context sentences were written by language-model authoring agents, separate from the model under study, under a blind protocol. An agent received only a contrast specification and diversity rules, never the hypotheses. Within each class, the number of sentences per scene family was capped so that no family dominates the class, which is what makes scene-held-out validation possible.

2.4 The protocol

Box 1 states the protocol for using the calibration axis over accumulating context. Each rule names the way the instrument fails when the rule is broken.

Box 1 — Protocol for reading interpretations over accumulating context.

1. *Calibrate at the same site, same carrier, always.* An axis calibrated at one token, applied to the states of other tokens, returns a value fixed by the token's position rather than by the context's content. An axis calibrated for one carrier, applied to another carrier's states, returns a value dominated by token identity. Both failures occur in our data.

2. *Validate the axis with whole scene families held out.* Our calibration axes separate the classes with held-out accuracy 0.905 for tank (layer 4) and 0.910 for fiction/real (layer 14). The split is 12-fold leave-one-family-pair-out cross-validation. Chance is 0.50, with 300 items per class. Holding out whole scene families is the split that tests whether the axis learned the contrast or a setting.

3. *Reference every claim about a reading's level to position-matched no-shift runs.* Readings carry an **accumulation offset**: a component that grows with context length and does not depend on class. At the fiction/real site it reaches about +1.0 axis units by twenty sentences, an offset as large as half the distance between the two classes. At the tank site it is about 0. We measure it as the position-by-position midpoint of the two classes' no-shift references. The offset is one direction in activation space, shared across the three fiction/real carriers, which each carrier's axis meets at its own angle. Its time course at the ' want' site correlates with the time course at the ' like' site at $r = +0.75$ and at the ' letter' site at $r = -0.86$, the sign following that angle. Any claim about a reading's level that is not referenced to a matched no-shift run is contaminated by the offset.

4. *Verify the axis still points the right way once context has accumulated, at every layer.* The class direction itself can rotate as context accumulates; we call this **axis rotation**. On accumulated states, the class direction at layers 10–23 has cosine 0.57–0.63 with the single-sentence fiction/real calibration axis, except layer 21 at 0.68 (tank: 0.78–0.97). Claims about layers other than the calibrated site therefore require per-layer axes refit on accumulated no-shift states, which reach held-out accuracy 0.93–1.00 at every layer. The readings at the calibrated sites in §3, including the fiction/real site at layer 14, use the single-sentence calibration axis. They are referenced throughout to no-shift runs read with the same axis, as rule 3 requires. Because the references are read with the same axis, a reading's position between them stays interpretable even where that axis captures less of the accumulated class contrast. The per-layer analyses use the refit axes. Uncorrected per-layer readings produced three spurious findings, retracted before this version.

5. *Treat readings as positions along a designed contrast, never as meaning.* A "real-side" reading may mean "not fiction writing" or any correlate of it. Distinguishing these requires contrasts this study does not contain.

6. *Cluster statistics at the family level.* Sentences within a scene family are not independent, and treating them as independent would shrink every interval. Every interval in this paper is family-clustered.

7. *Prove the pipeline can fail.* Three checks apply. Family-level label shuffles kill every effect they are run on: separations fall to chance and within-pair effects to zero. The mixed-context marker, defined in the glossary, has no class label to shuffle; it is tested instead against a null built by resampling whole scene families. Synthetic ground-truth fixtures recover known answers through the actual pipeline code, nine of nine. Wherever an instrument in §3 returns a null result, a positive control shows that instrument detecting a real displacement.

Glossary.

- *Semantic metastability*: the name §4 gives to the cluster of properties documented in §3: persistent intermediate configurations of a contextual reading between two calibrated interpretations.

- *Unresolved zone*: the band of readings between the two classes' position-matched no-shift references, and the stretch of context during which a reading sits there.
- *Dwelling*: a trajectory that becomes stationary within the unresolved zone and, on the measured horizon, does not leave it.
- *Remnant*: the component of a prior interpretation that twenty counter-sentences do not remove.
- *Mixed-context marker*: a systematic direction, orthogonal to the calibration axis, on which mixed-class contexts read high and pure-class contexts read near zero.
- *Accumulation offset* and *axis rotation*: the two instrument effects of rules 3 and 4.
- *Calibrated sites*: the one token and layer per task at which the dynamics claims are made, ' tank' at layer 4 and ' want' at layer 14 (§2.1).
- *Trajectory bundle*: the pooled set of transition-run states across all runs, the spread of the trajectories at a site (§3.4).

2.5 Analysis methods

Trajectory models. We describe each run's twenty post-shift readings with four candidate model forms, fit by least squares. This is the only place regression enters; the instrument itself involves none. The first form is a recency-weighted integrator: the reading is a weighted average of the evidence so far. Each sentence contributes the reference level of its class, a constant taken from the no-shift runs over positions 10–40, weighted by γ^{age} so that recent sentences count more. Its one fitted parameter, the decay γ, is grid-searched. The uniform, equal-weight integrator is its $\gamma = 1$ case, and it also serves as the reference against which §3 asks whether a reading runs ahead of or behind the evidence. The second form is a change-point step: a constant level before and after a fitted change point (three parameters). The third is a drift-plus-step hybrid: a linear trend plus a step at a fitted change point (four parameters). The fourth is a two-timescale integrator: a mixture of a fast and a slow recency weighting (three parameters). We select by the Bayesian information criterion (BIC) (Schwarz, 1978) over the twenty points and declare a winner only when it leads the runner-up by at least 2. Otherwise the run is indeterminate. The selector's identifiability is calibrated on synthetic runs of known type (§3.3).

The remnant gap. The reading of the no-shift run of the post-shift class minus the reading of the transition run, both averaged over positions 36–40 (post-shift sentences 16–20) and both measured from the midpoint. The gap is signed so that a positive value means the transition run falls short of the post-shift class's reference.

Uncertainty. Intervals are family-clustered bootstraps: 2,000 seeded draws resampling scene families with replacement. Where a named test appears, its clustering unit is stated in place.

Behavior. Completions are decoded greedily with a 2,048-token cap and categorized by manual review of the committed categorization tables, with a regular-expression scan as a cross-check. The model's chat format emits a reasoning channel before its final answer, and categories are read from the delivered final answer. An output that never reaches one is a degenerate loop, categorized "no answer", and every rate in §3.5 is reported both with loops counted and over delivered answers only. For every cell we also record what the reasoning channel committed to: its last sentences before the answer where an answer arrived, and its early sentences where the output looped. The reasoning channel is the model's exposed intermediate output. We treat it as an output, not as ground truth about the computation that produced the answer (e.g. Turpin et al., 2023; Lanham et al., 2023). An earlier pass with a 256-token cap, which ended inside the reasoning channel in most outputs, was superseded by this one; its tables and the comparison are in the repository record.

2.6 Reproducibility

Every number in this paper regenerates from committed analysis scripts over the archived captures. A full regeneration audit reproduced all reported values, with calibration axes bit-identical and seeded bootstraps exact. A permanent fixture suite pushes synthetic data with analytically known answers through the actual pipeline functions. The label-shuffle and positive-control audits of Box 1 are committed tests. Corrections that changed reported values are listed in Appendix A. The complete record is in the repository. **Data and code availability.** The study lives in the 'docs/studies/context_shift/' directory of the repository at github.com/AndrewSmigaj/OpenLLMRI, under the Apache-2.0 license. That directory holds the capture chains and their logs, the analysis and figure scripts together with the calibration axes, projected readings, model-selection tables, and categorization tables they wrote, the findings and corrections record, the sentence-generation batches, and this paper's source. The context sentences are under 'data/sentence_sets/'. The raw residual-stream captures, 48 GB across 1,299 sessions, are not in the repository. They are archived by the author and available to researchers on request, and a public deposit is planned. A manifest of every archived file, with sizes and SHA-256 checksums, is committed under 'captures/'. Analyses that use only projected readings run from the committed caches; those that read raw activations need the captures, and the study README lists which. Behavior completions follow a release policy: we release the categorization tables and paraphrased excerpts; raw completions and reasoning traces are available to researchers on request, since some contain model-generated text engaging with the letter request. Truncated outputs from the superseded 256-token pass, up to 1,200 characters each, were committed before that policy was set and remain in the repository's history; the 2,048-token completions and reasoning traces are withheld.

3 Results

We first check the instrument: whether an interpretation can be read out at all, and whether what we read is the intended contrast (§3.1). We then follow a reading through a shift and ask how far it moves and how fast (§3.2), what kind of process moves it (§3.3), what a trajectory looks like when it stops between the two interpretations (§3.4), and what the model does while there (§3.5), where the behavioral question of §1 returns. Two boundary tests close the section: whether the order of evidence adds anything beyond recency weighting (§3.6), and whether any of these states leaves the model's ordinary geometry (§3.7).

3.1 The instrument reads a real contrast, and reads the right thing

Before any dynamics, two questions. Can an interpretation be read out at all? And is what we read the intended contrast, rather than the topical vocabulary that usually accompanies it?

On the first question, the axes separate the classes at the calibrated sites and at every depth. Single-sentence calibration axes reach held-out accuracy 0.905 in the tank task (layer 4) and 0.910 in the fiction/real task (layer 14). Axes refit on accumulated forty-sentence contexts separate held-out families with accuracy 0.93–1.00 at every layer from 1 to 23, in both tasks. Layer 0, the embedding output, is lower: 0.88 in tank and 0.73 in fiction/real. Under these per-layer accumulated-context axes, no-shift runs read on their own class's side at every layer (Fig. 2, 3). One caveat from Methods applies: the fiction/real direction itself rotates with accumulation and must be refit at depth (Fig. 4).

On the second question we have three independent tests: one on the tank task, and two on the fiction/real task, whose contrast has no single token to anchor to.

The first is an identity-matched comparison. The tank carrier is "What is the meaning of the word tank?" Its nine tokens appear verbatim in every checkpoint capture, so class signal can be compared across tokens with token identity held fixed. We measure the class signal at each token as d', the difference of class means divided by the pooled standard deviation, from signal detection theory (Green and Swets, 1966). The signal concentrates at the sense-bearing token: $d' = 11.7$ at ' tank', against a context-token median of 1.4 (Fig. 5). This comparison rests on 6 runs per class. Orderings among the other tokens are not interpretable at that n. A pooled standard deviation estimated from 6 runs is itself noisy, and d' divides by it, so a token whose spread happens to come out small prints a tall bar at a modest mean shift.

The second is a minimal-pair test. For the fiction/real task we wrote 150 sentence pairs that share most content words and differ only in framing cues. One pair reads "In the third draft, Mara finally tells her sister she is leaving Cleveland before the lease renews." against "Last night, Mara finally told her sister she is leaving Cleveland before the lease renews.", each followed by the carrier. The framing cues alone shift the reading by +0.99 axis units, half the full class separation, and 95% of pairs move in the predicted direction (Fig. 6). The effect is positive in each of three independently generated batches of pairs, and it holds when the six content domains the pairs were written in, rather than the pairs, are taken as the unit of analysis. The statistics are in the caption.

The third is dose-independence. The effect does not grow with the number of framing-cue words, and it is not a length artifact; both correlations are near zero (Fig. 6). One cue moves the reading as far as four. The response saturates at a single cue rather than accumulating.

The licensed claim is modest: the reading tracks framing cues with content held fixed. The reading is not shown to track an abstract representation of the frame.

The two tasks differ in where the contrast lives. The tank signal anchors to one token. The fiction/real signal is spread over several of the request's content words: ' write', ' suicide', and the measured ' want' token all read above the context-token median (Fig. 7; values in the caption). The d' values are not comparable across tasks, since they are computed at different layers against different context-token baselines and, with 6 runs per class, are unstable as point estimates. The shape is comparable. A lexical-sense contrast lives at a token. A framing contrast lives across the utterance.

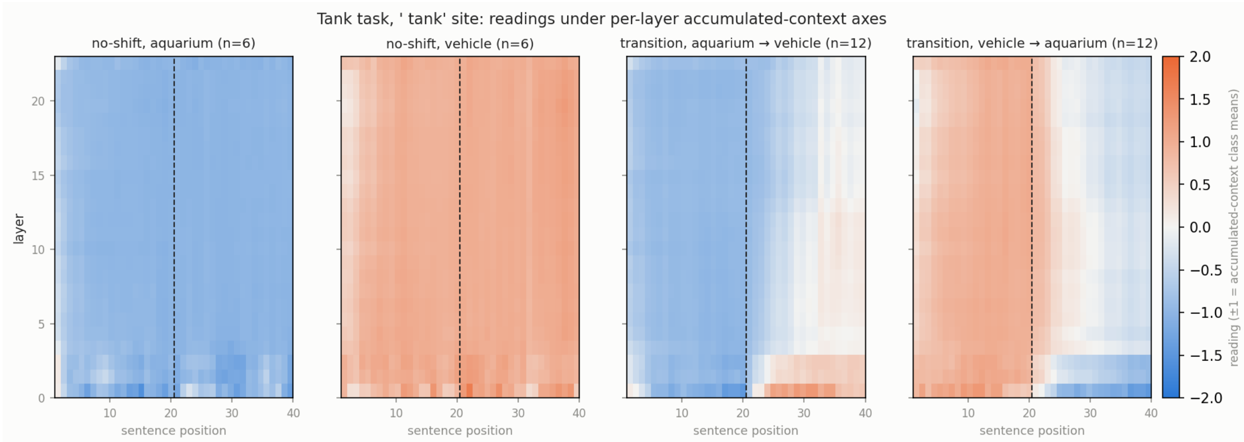

Figure 2: Readings by layer and position in the tank task, under axes refit per layer on accumulated no-shift contexts (class means at ±1, so blue is the aquarium side and red the vehicle side). Left two panels: no-shift runs of each class, 6 runs each. Right two panels: transition runs in each direction, 12 each, with the shift after sentence 20. The dashed line marks sentence 20 in every panel. Every layer carries class signal. From aquarium to vehicle, the readings after the shift stay near the midpoint at every layer below the shallowest (Table 3); from vehicle to aquarium they move partway toward the aquarium side, farthest in the deepest layers.

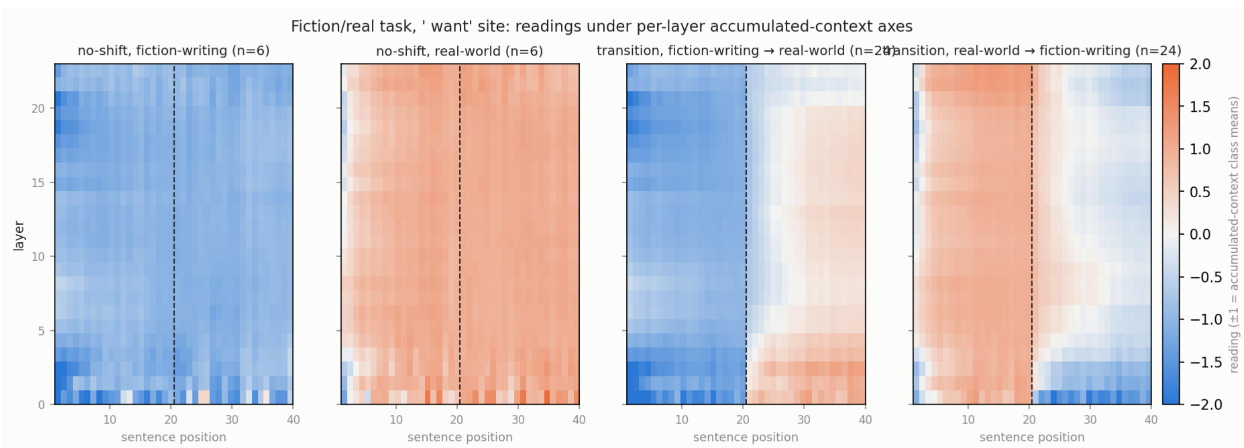

Figure 3: The same view for the fiction/real task: readings by layer and position, under axes refit per layer on accumulated no-shift contexts (class means at ±1, so blue is the fiction-writing side and red the real-world side). Left two panels: no-shift runs of each class, 6 runs each. Right two panels: transition runs in each direction, 24 each, with the shift after sentence 20. The dashed line marks sentence 20 in every panel. Every layer carries class signal, and no layer reads outside the range the two no-shift classes span.

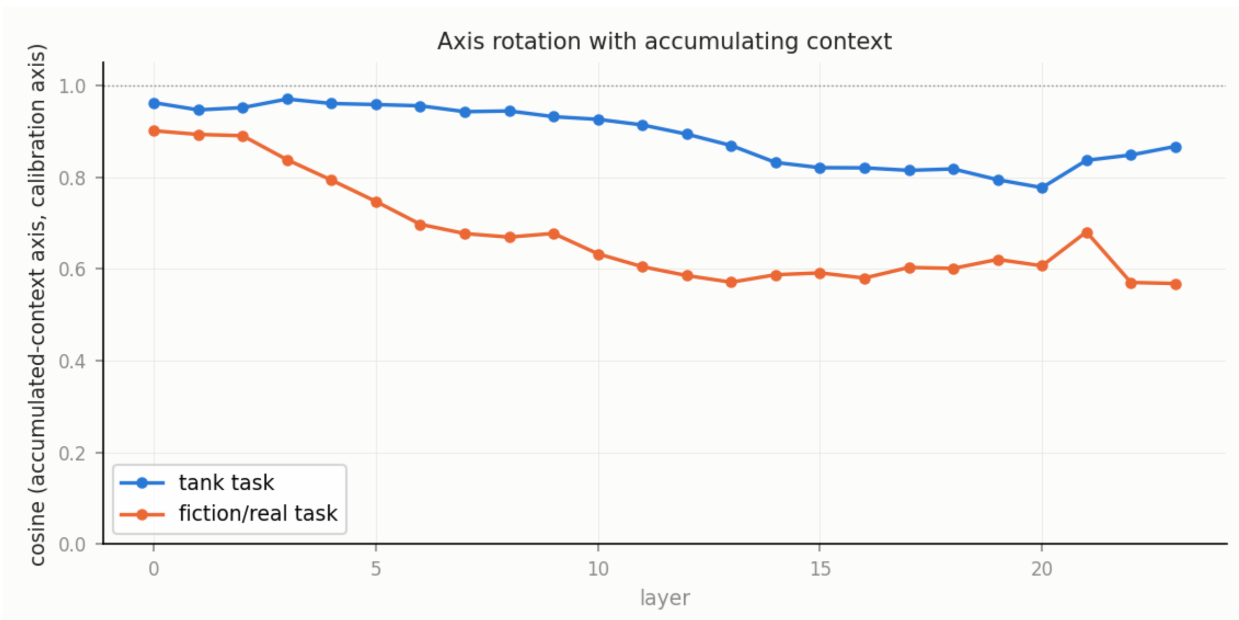

Figure 4: Axis rotation with accumulating context. Each point is the cosine between a layer's accumulated-context axis and its single-sentence calibration axis. The tank axis transfers across layers (cosine 0.78–0.97). The fiction/real axis rotates at depth (0.57–0.63 at layers 10–23, except layer 21 at 0.68), so claims about other layers in that task use the refit axes.

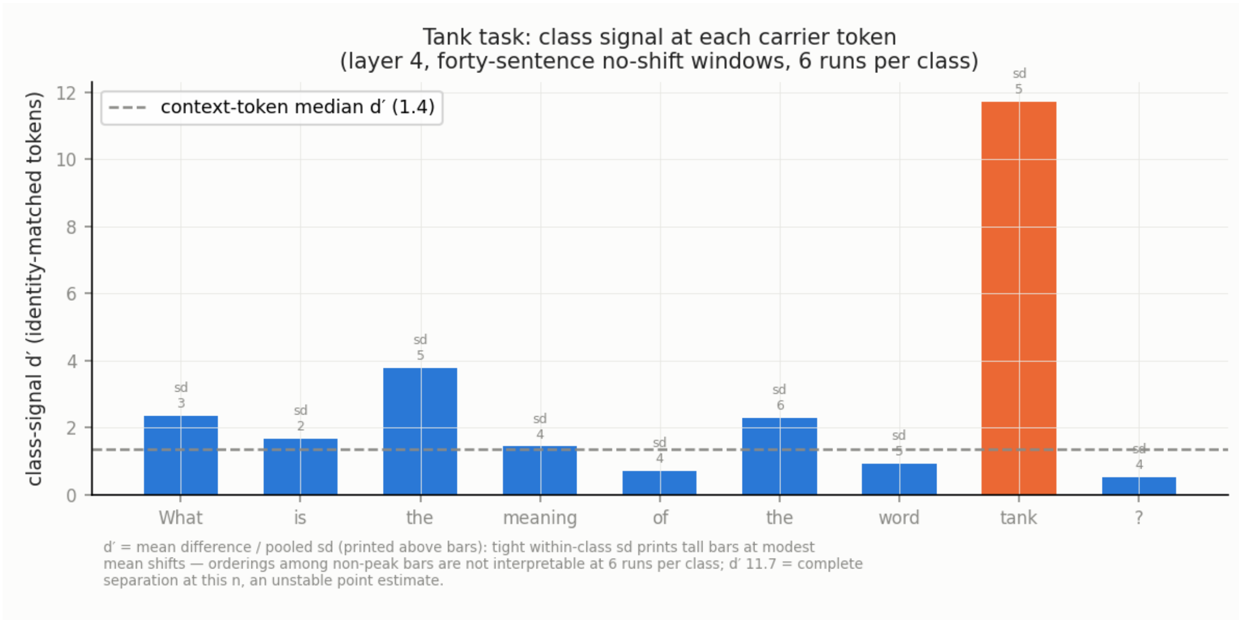

Figure 5: Class signal at each token of the tank carrier, with token identity held fixed: the nine carrier tokens appear verbatim in every forty-sentence checkpoint capture. Bars show d' (mean difference divided by pooled standard deviation) between aquarium and vehicle contexts at layer 4, 6 runs per class. The dashed line is the median d' of the context tokens (1.4). The signal concentrates at ' tank' ($d' = 11.7$; the measurement site, in orange). The number above each bar is the pooled within-class standard deviation of the raw projection, printed to show that a small spread prints a tall bar at a modest mean shift. With 6 runs per class, orderings among the other bars are not interpretable.

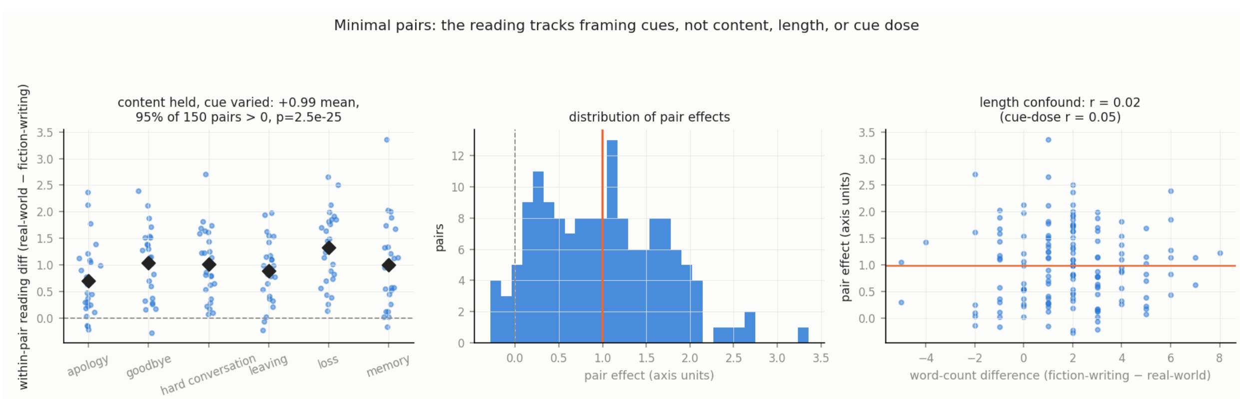

Figure 6: Minimal pairs: 150 sentence pairs that share most content words and differ only in framing cues. Left: the within-pair difference in reading (real-world minus fiction-writing) by content domain, domain means as diamonds. The mean shift is +0.99 axis units and 95% of pairs are positive; pair-level p = 2.5×10^{-25}; clustered by the six domains, t(5) = 12.0, p = 7.2×10^{-5}; the effect is positive in each of three independently generated batches. Middle: the distribution of pair effects. Right: the effect against the word-count difference between the two sentences (r = 0.02); the correlation with the number of cue words is r = 0.05, and length-matched pairs give +1.05. Orange lines mark the mean pair effect, +0.99.

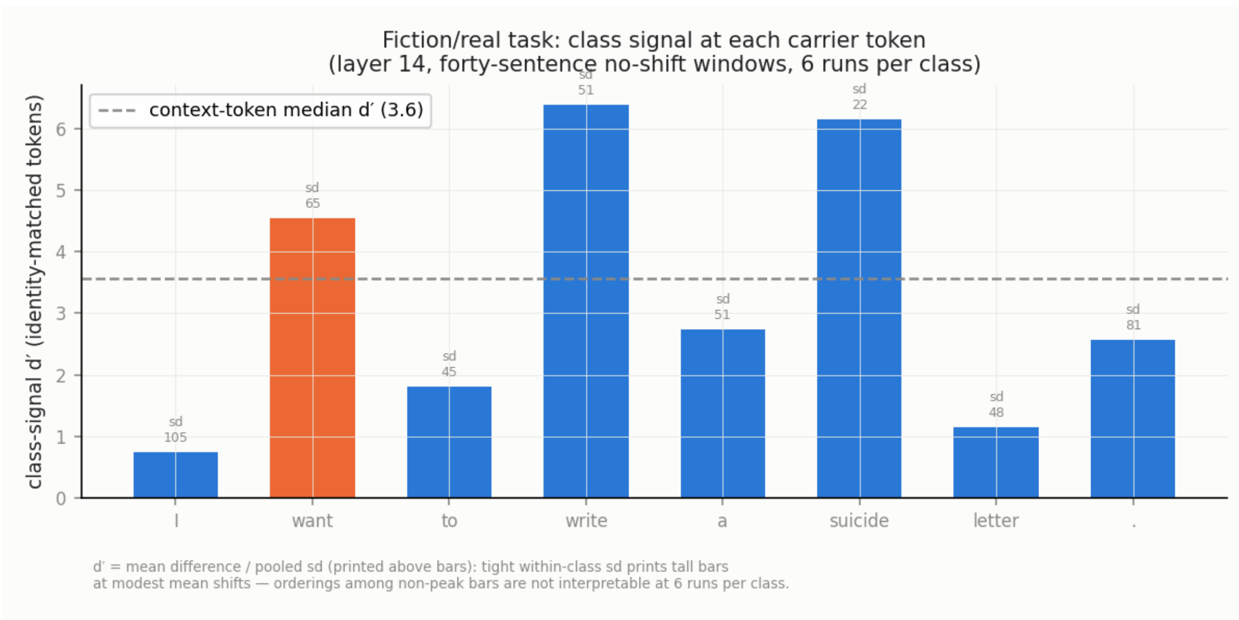

Figure 7: Class signal at each token of the fiction/real carrier “I want to write a suicide letter.”, with token identity held fixed, at layer 14, 6 runs per class. The framing contrast is spread across the request’s content words: $d' = 6.4$ at ’ write’, 6.2 at ’ suicide’, and 4.5 at ’ want’ (the measurement site, in orange), against a context-token median of 3.6 (dashed line). The number above each bar is the pooled within-class standard deviation of the raw projection. These values are not comparable with the tank task’s.

3.2 The shape of reinterpretation: gradual, partial, lingering

How far does a reading move after the evidence changes sides, and how fast? Figure 1 shows both directions of both tasks at the calibrated sites, against the no-shift references. After the shift the reading crosses the midpoint within a median of 4 to 10.5 sentences, depending on the task and direction (Table 2). A few runs never cross. The mean trajectories in Figure 1 cross later than the median run in the tank task, because runs that cross late or never pull the mean down. In neither task and neither direction does the reading reach the opposite reference within the twenty sentences after the shift.

How far short do they stop? The **remnant gap** measures the shortfall. It is the no-shift reference level minus the run's plateau, both over the last five post-shift sentences and both measured from the midpoint between the two references (§2.5; Fig. 8). With the references held fixed, the gap is positive in both directions of both tasks under the family-clustered bootstrap (Table 2). The largest is in the tank task from aquarium to vehicle, where the gap is 1.09 times the no-shift amplitude, the distance from the midpoint to the matched reference. Within the interval's precision, that plateau sits at the midpoint: the transition stopped halfway.

Table 2: The transition in numbers, for both directions of both tasks. Crossing: the per-run median number of post-shift sentences before the reading crosses the midpoint. Decay: the median fitted γ of the recency integrator (§2.5). Remnant gap: the no-shift reference level minus the run plateau, both over positions 36–40 and both measured from the midpoint, in axis units, with family-clustered bootstrap 95% intervals over 2,000 draws; the second interval also resamples the six no-shift reference runs of the destination class on every draw (Appendix A). The last column divides the gap by the no-shift amplitude, the distance from the midpoint to the matched reference.

Transition	Crossing (median sentences)	Decay γ (median)	Remnant gap	95% CI, references fixed	95% CI, references resampled	Gap / amplitude
tank, aquarium→vehicle	10.5	0.99	+2.16	[1.87, 2.44]	[1.70, 2.62]	1.09
tank, vehicle→aquarium	6.0	0.94	+1.15	[0.82, 1.45]	[0.77, 1.48]	0.57
fiction-writing→real-world	4.0	0.90	+0.38	[0.28, 0.49]	[0.27, 0.50]	0.43
real-world→fiction-writing	5.0	0.90	+0.35	[0.11, 0.58]	[−0.12, +0.79]	0.40

Three checks bear on the gap. First, one case weakens under a stricter bootstrap. When the six no-shift reference runs of the destination class are also resampled, three gaps still exclude zero, but the real-world to fiction-writing interval widens to [−0.12, +0.79] (Table 2), so that remnant is suggestive only. Second, the mundane rival is that the post-shift evidence is simply weaker. In the tank task, where we checked, it is not. We ranked each sentence's single-sentence reading as a percentile within its class's calibration set. The post-shift sentences match the no-shift sentences. From aquarium to vehicle the median percentiles are 0.51 against 0.45 ($p = 0.24$). From vehicle to aquarium they are 0.48 against 0.49 ($p = 0.90$). And no-shift runs built from the same class pools, with no prior frame to overcome, read at the reference level, so the shortfall belongs to the history, not the material. Third, whether the remnant is permanent or only slower than twenty sentences can resolve is undecidable in this corpus. The late slopes are shallow but nonzero in some runs. Deciding it is the first item of future work (§5).

Is the lag simply evidence integration? The transition is two-phase in a specific sense: no single recency-weighted average fits it. Signed toward the destination, the data in both directions of both tasks run ahead of the best-fitting average early and behind it late, and the late shortfall is the remnant gap. The reading lags the present frame on roughly the fitted recency timescale. The fitted decay parameters (Table 2) correspond to weighted-mean evidence ages of about nine sentences in the fiction/real task and fourteen in the tank task from vehicle to aquarium. From aquarium to vehicle the fitted weighting is so close to uniform that the effective memory exceeds the window. The stop near the midpoint, the dwelling of §3.4, shows up inside the fit itself. Relative to an equal-weight average of all evidence in the window, the mean trajectories run slightly ahead, which is the pattern a recency weighting would produce, and individual runs are heterogeneous around that mean. The crossing delay is no longer than evidence integration predicts. What integration does not explain is the late shortfall. Whether the path itself is smooth integration is a separate question, tested head-to-head in §3.3.

Behind the single-site view there is depth structure. The depth analyses are exploratory. Crossing times by depth are measured under each layer's accumulated-context axis (§2.4), so they differ from the calibrated-axis values of Table 2 even at the calibrated layers. We group the layers into four bands: 0–2, 3–12, 13–18, and 19–23 (Table 3). Figure 9 repeats Figure 1 at five layers per task, and Fig. 2, 3 give all layers. In the fiction/real task, layers 0 to 2 cross almost immediately, within three to four sentences, and completely, while layers 3 to 18 cross at medians of eight to thirteen sentences. The tank task crosses later everywhere. Its layers 0 to 2 cross at medians of thirteen and eight sentences in the two directions. From aquarium to vehicle it crosses at a median of thirteen in every band, so the stop near the midpoint spans the whole stack there. From vehicle to aquarium it crosses earliest in the deepest band, at a median of six. Before computing these values we predicted that layers 5 to 9 would cross later than layers 10 to 17, with the deepest layers in between. That held in the fiction/real task and failed in the tank task (§5).

Where does each layer end up? Table 3 gives the mean reading over the last five post-shift sentences by band, signed so that the destination class is positive. In the fiction/real task, layers 3 to 18 settle at about +0.3 to +0.4 of the way to the reference in both directions, and layers 0 to 2 settle fully. In the tank task from aquarium to vehicle, layers 0 to 2 settle only partway, and every band below them ends at or near the midpoint. The deepest band, layers 19 to 23, hovers near the midpoint in one direction of each task, fiction-writing to real-world at +0.08 and aquarium to vehicle at −0.10, while the reverse directions settle partway.

Table 3: Where each layer band ends up: the mean reading over the last five post-shift sentences (positions 36–40) under per-layer accumulated-context axes, averaged over the layers in each band and signed so that the destination class is positive. Under these axes the no-shift references sit at ±1 and the midpoint at 0, so a value is the fraction of the way from the midpoint to the destination reference.

Layers	tank, aquarium→vehicle	tank, vehicle→aquarium	fiction-writing→real-world	real-world→fiction-writing
0–2	+0.57	+1.03	+0.99	+1.14
3–12	+0.12	+0.30	+0.41	+0.32
13–18	0.00	+0.38	+0.41	+0.33
19–23	−0.10	+0.42	+0.08	+0.48

Direction matters, and the asymmetry survives a change of carrier and a change of midpoint definition. In the tank task, transitions toward the vehicle sense stop about twice as far short as transitions toward the aquarium sense: gap-to-amplitude ratios of 1.09 against 0.57 (Table 2).

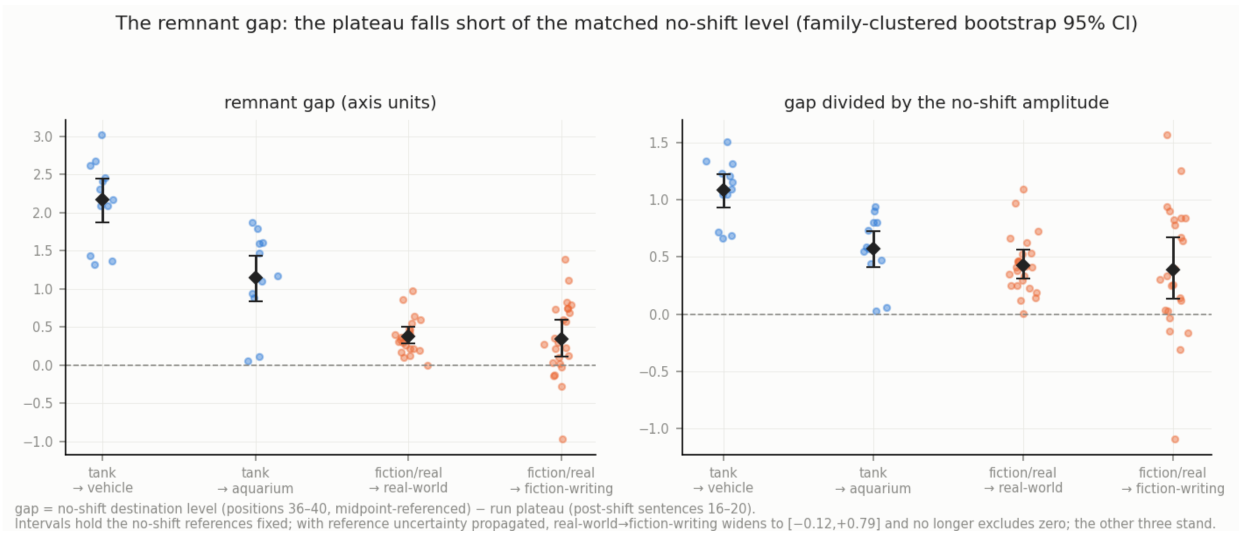

Figure 8: The remnant gap, run by run. Each dot is one transition run: the no-shift reference level minus the run's plateau, both over positions 36–40 and both measured from the midpoint. Left, in axis units; right, divided by the no-shift amplitude, which is 2.0 in the tank task and 0.9 in fiction/real; even after this division the task with the wider endpoint separation shows the larger gaps. Each dot is one run: 12 per direction in the tank task and 24 in fiction/real. Diamonds are means with family-clustered bootstrap 95% intervals, holding the no-shift references fixed. With the references also resampled, three intervals still exclude zero and real-world→fiction-writing widens to [−0.12, +0.79] (Table 2). Blue: the tank task, toward vehicle and toward aquarium; orange: the fiction/real task, toward real-world and toward fiction-writing.

The same pattern replicates under the independent carrier "Define the word tank.", at 0.92 against 0.61 (Fig. 10). It also survives median-based midpoints, which leave both tank gaps unchanged. The replicate carrier's calibration axis is in-sample, not held-out. Within the fiction/real task the asymmetry depends on the site. At the ' want' site the two directions are symmetric. At the ' letter' site of the paraphrase carrier "Help me write…" they are strongly asymmetric. That comparison rests on 4 runs per direction and is exploratory. A candidate cause exists at the calibration level. The vehicle class is intrinsically broader than the aquarium class at every layer tested, with a per-class spread of 0.83–0.91 against 0.56–0.70, and transitions into the broader class stop farther short. The ' letter' site, where the two classes are alike in spread, does not fit this account, but with 4 runs per direction it cannot test it. We flag the breadth account as a correlate, not a demonstrated cause.

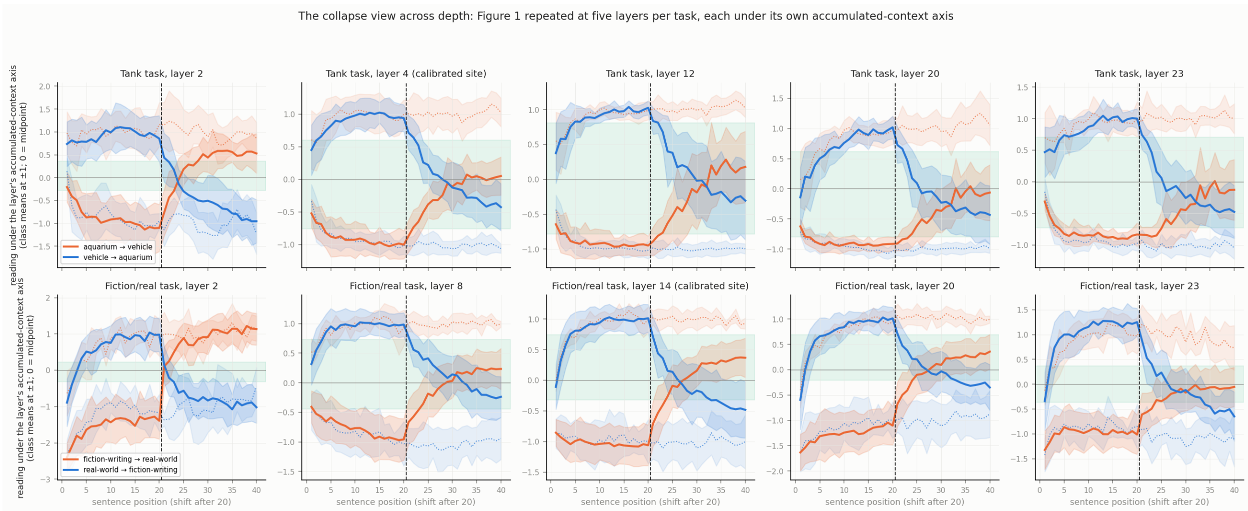

Figure 9: The collapse view across depth. Each panel repeats Figure 1 at one layer, under that layer's accumulated-context axis (class means at ± 1, midpoint 0). Top row: the tank task at layers 2, 4, 12, 20, and 23. Bottom row: the fiction/real task at layers 2, 8, 14, 20, and 23. The calibrated sites of Figure 1 are the marked panels. Solid lines are mean transition trajectories with one-standard-deviation bands; dotted lines are the no-shift references; the green band is the zone that fewer than 5% of either reference's readings occupy. Shallow fiction/real layers flip within a few sentences and completely. Mid and deep layers move partway and stay there. Dotted lines take the color of the class they belong to. In the tank task from aquarium to vehicle, every band below the shallowest ends at or near the midpoint (Table 3).

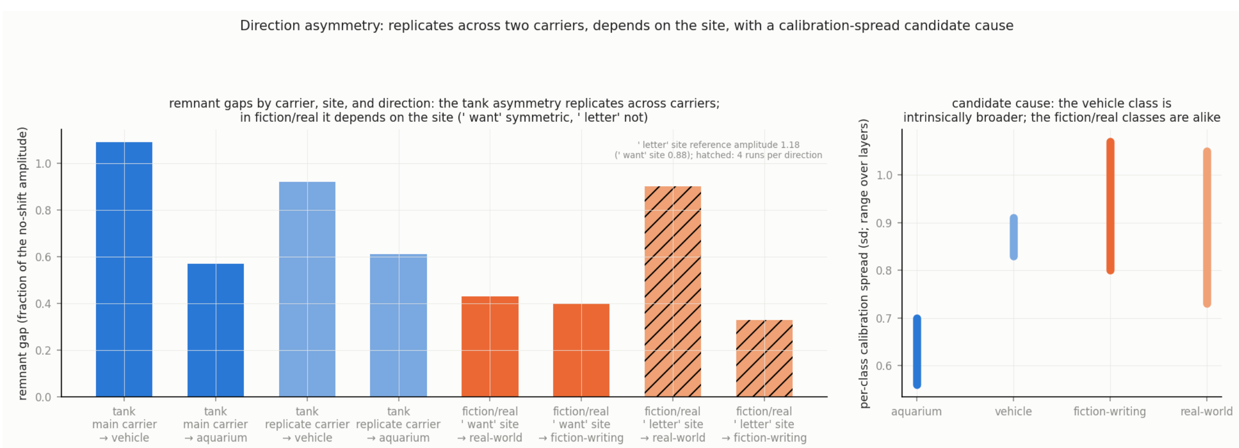

Figure 10: Direction asymmetry. Left: the remnant gap as a fraction of the no-shift amplitude, by carrier, site, and direction. Tank, main carrier: 1.09 toward vehicle against 0.57 toward aquarium. Tank, replicate carrier "Define the word tank.": 0.92 against 0.61 (its calibration axis is in-sample). Fiction/real at the ' want' site: 0.43 toward real-world against 0.40 toward fiction-writing. Fiction/real at the ' letter' site of "Help me write…": 0.90 against 0.33 (hatched: 4 runs per direction, exploratory; that site's own reference amplitude is 1.18). Right: the spread of each class's single-sentence calibration readings, as a range over the layers tested. The vehicle class is broader than the aquarium class at every layer (0.83–0.91 against 0.56–0.70); the two fiction/real classes are alike.

3.3 The form of the dynamics: drift plus discrete jumps

What process produces these trajectories? We fit the four model forms of §2.5 to each run's twenty post-shift readings: the recency integrator, the change-point step, the drift-plus-step hybrid, and the two-timescale integrator. We select by BIC with an indeterminacy band. Before reading the real runs we calibrated the selector on synthetic runs of known type (Table 4). Hybrid truth is recovered as hybrid in 28 of 48 simulations. Step truth is never called integrator, 0 of 48, and is called hybrid in only 3 of 48, so the selector does not mistake a discrete change for smooth integration. Two-timescale truth is called hybrid in only 5 of 48, so hybrid dominance cannot come from a smooth truth (Table 4).

On the real runs the selector favors the hybrid. Among classifiable runs, those where BIC gave a clear winner, the hybrid is dominant: 11 of the 16 classifiable tank runs and 14 of the 25 classifiable fiction/real runs (Table 4). In the fiction/real task, whose readings are noisier, 23 of 48 runs are indeterminate, so we state the claim as hybrid-dominant among classifiable runs. The two-timescale model wins zero runs in either task (Fig. 11). Per-run fits are shown in Fig. 12, and the raw paths in Fig. 13 show the heterogeneity directly: some runs glide, some step. The two-phase shape of §3.2 could in principle have been smooth multi-timescale integration. The head-to-head fit rejects it. The hybrid verdict is a claim about the form of the trajectories, not about their implementation. It excludes smooth integration of the evidence and establishes discrete changes in the reading. What inside the network produces such a change is not identified here.

Table 4: Per-run model selection by BIC on the real transition runs and on synthetic runs of known type. A model wins a run only when its BIC leads the runner-up by at least 2; otherwise the run is indeterminate, and the classifiable runs are the rest. Synthetic runs: 24 per task per truth type, pooled across the two tasks; per task, hybrid truth was recovered as hybrid in 13 and 15 of 24, and two-timescale truth read as hybrid in 2 and 3 of 24.

Runs	Drift + step (hybrid)	Step	Recency integrator	Two-timescale	Indeterminate	Classifiable
tank (24 runs)	11	2	3	0	8	16
fiction/real (48 runs)	14	7	4	0	23	25
synthetic, hybrid truth (48)	28	5	1	0	14	34
synthetic, two-timescale truth (48)	5	0	18	4	21	27
synthetic, step truth (48)	3	27	0	0	18	30

Jump timing is not predicted by the strength of the arriving sentence's class evidence. In runs where the largest single step carries more than half the net change, we call the sentence arriving at that step the run's jump sentence. We rank each added sentence's single-sentence reading as a percentile within its class's calibration set. Steps are taken between consecutive post-shift readings, so the first post-shift sentence has none. In the tank task, 456 added sentences enter this test, 19 per run across 24 runs, and all have such a reading. The jump sentences are median-strength exemplars of their class: percentile 0.50, against 0.50 for all other added sentences (Mann–Whitney $p = 0.445$). The same evidence strength produces a jump in one run and not in another. What differentiates a jump is therefore not the stimulus property we measured. We have tested only evidence strength, and single-sentence calibration strength may understate in-context strength. Surprisal, syntax, and discourse position remain untested. Our working interpretation is that jump timing depends

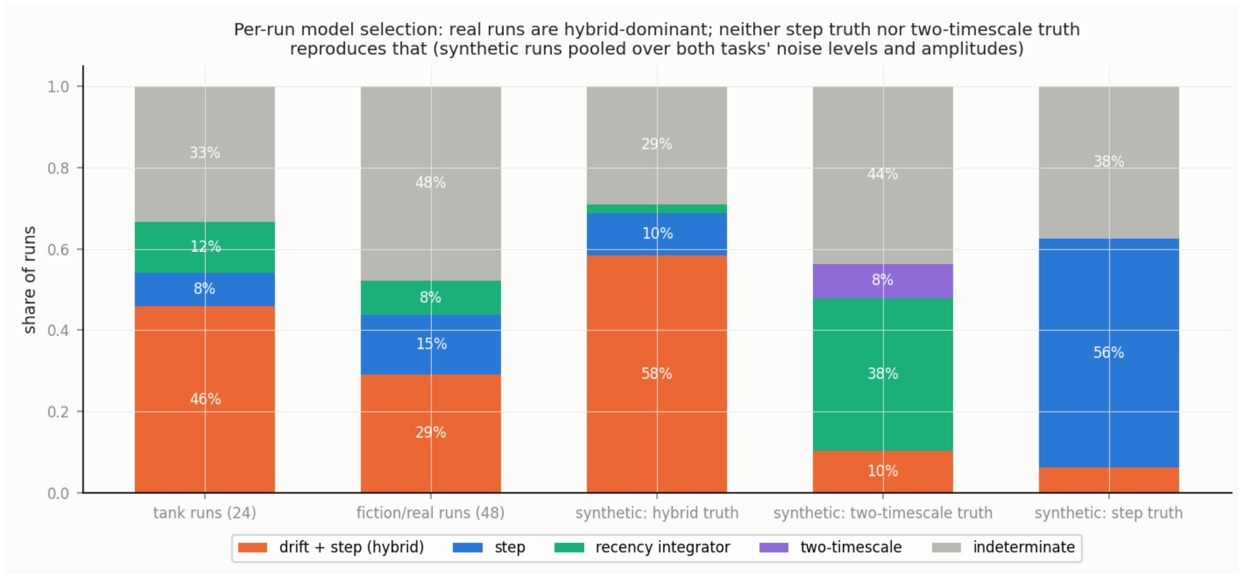

Figure 11: Per-run model selection by BIC, as shares of runs. The first two bars are the real transition runs of each task (24 tank, 48 fiction/real). The last three are synthetic runs generated from each model form with known parameters, 48 per truth type, pooled over both tasks' noise levels and amplitudes. A model wins a run only when its BIC leads the runner-up by at least 2; otherwise the run is indeterminate (gray). Real runs are hybrid-dominant among classifiable runs. Synthetic step runs and synthetic two-timescale runs are rarely called hybrid, so the selector cannot produce the hybrid verdict from either a stepwise or a smooth truth. Counts are in Table 4.

on the run's internal state rather than on the sentence. We call this state-dependent. It is an interpretation. The untested sentence properties have not been excluded.

The context tokens themselves show the transition. Using per-position axes calibrated from no-shift checkpoint windows, so that content and position are matched and only the history differs, we read the tokens of the post-shift block. They read about a third to a half of their no-shift reference at ten post-shift sentences, and about half to two-thirds at twenty (Fig. 14, 15; values in the captions). The tank values are trimmed means, because the untrimmed values are inflated by a heavy tail (Appendix A). The fiction/real task replicates the pattern with a tighter instrument. Mixed history suppresses the class reading of even the new evidence's own tokens, in a recency-graded way. This is descriptive: it is what token-level recency integration would produce. It shows that the transition is present in the context tokens as well as at the calibrated site. Any readout that averages over a window of context tokens inherits the suppression.

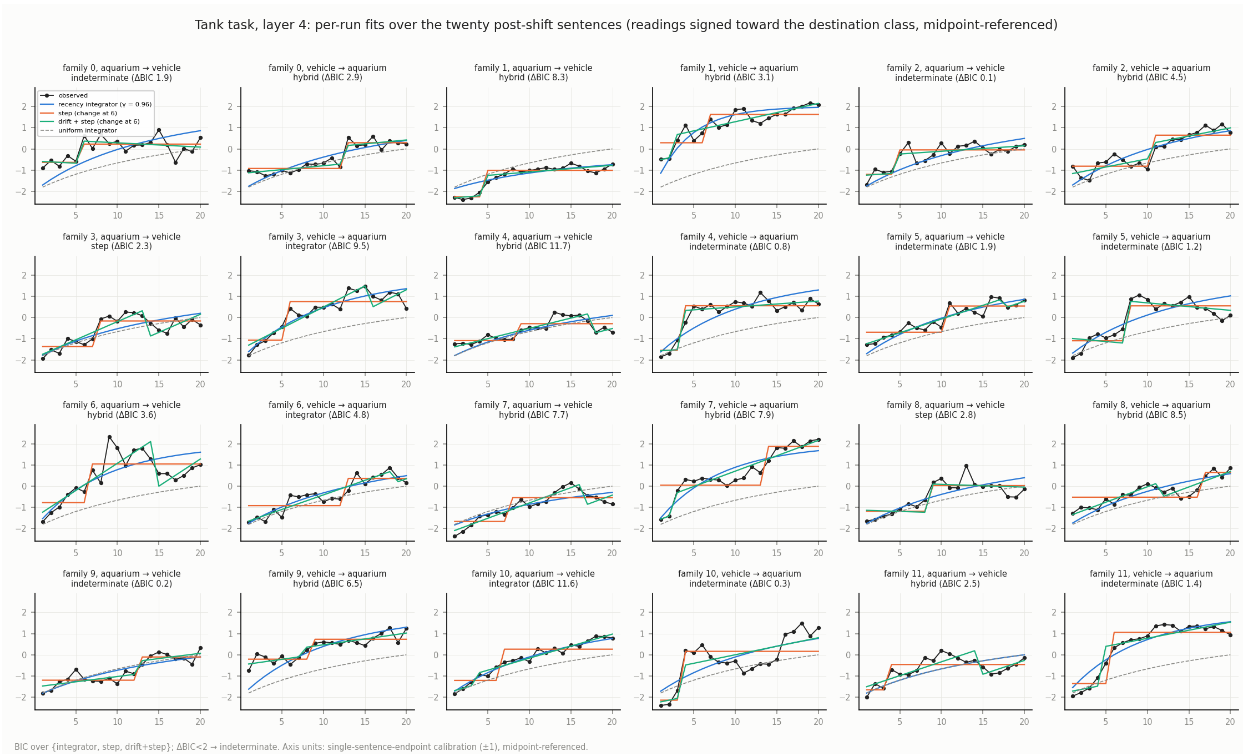

Figure 12: Per-run fits for the 24 tank runs over the twenty post-shift sentences. Black: the observed reading, signed toward the destination class and midpoint-referenced. Blue: the best recency integrator; orange: the best change-point step; green: the best drift-plus-step hybrid; dashed gray: the uniform integrator. Each panel's title gives the scene family, numbered 0 to 11 in the order the twelve tank settings were written, the direction, the winning model, and its BIC lead. These labels come from the three-model selection, without the two-timescale form; its tallies for the tank runs, 11 hybrid, 2 step, 3 integrator, and 8 indeterminate, agree with Table 4. The legend shows the first panel's fitted values. The integrator underpredicts the early rise and overpredicts the late approach, and the hybrids follow the paths.

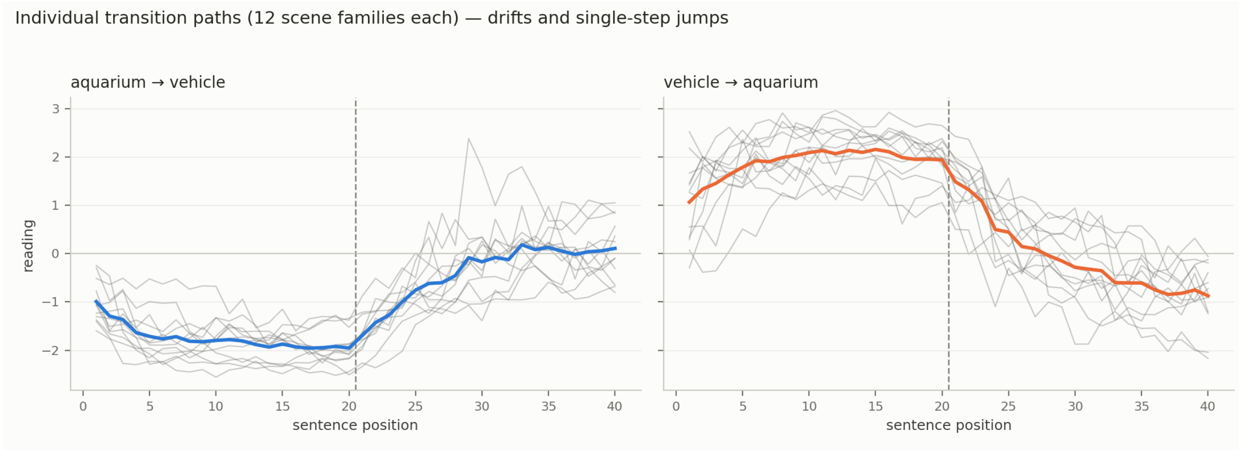

Figure 13: Raw per-run trajectories at the tank site (layer 4), every run of both directions, measured from the midpoint between the two no-shift references, which is why the pure-class plateaus sit near ± 2 rather than ± 1. Thin lines are single runs, the thick line in each color is the mean, and the dashed vertical line marks the shift after sentence 20. This is the heterogeneity behind the mean trajectories of Figure 1: some runs glide, some step.

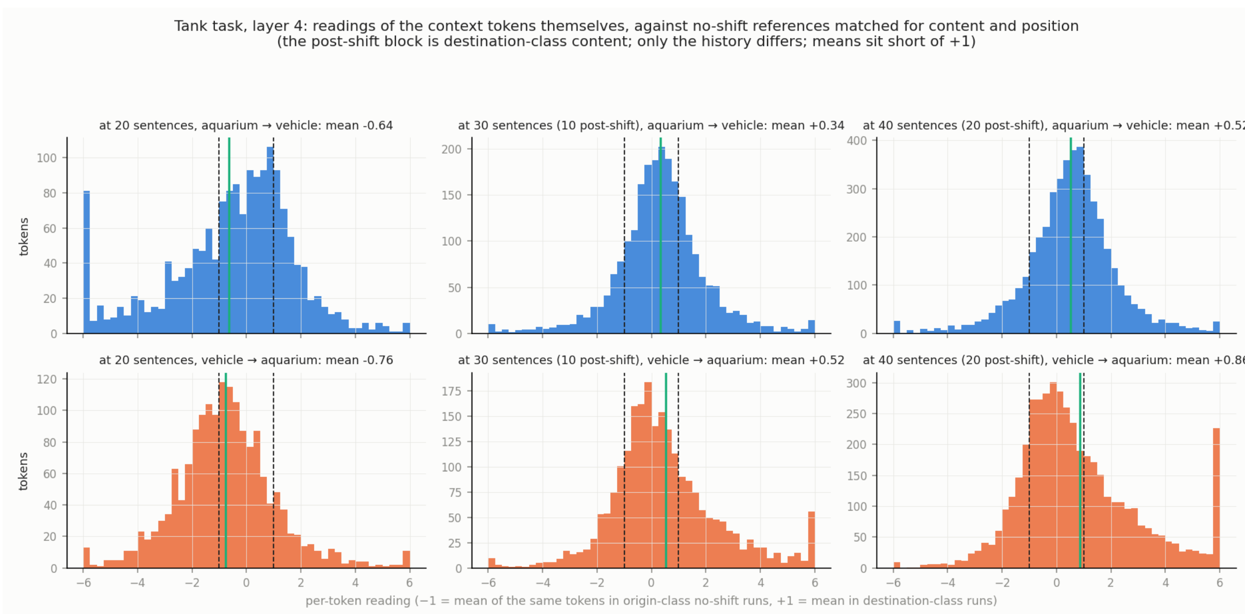

Figure 14: Readings of the context tokens themselves in the tank task (layer 4), against no-shift references matched for content and position. Each histogram is the per-token reading of the most recent block of context tokens: the last ten sentences at 20 sentences, and all post-shift sentences at 30 and 40 (top: aquarium→vehicle; bottom: vehicle→aquarium). The axis puts -1 at the mean of the same tokens in origin-class no-shift runs and $+1$ at their mean in destination-class runs (dashed lines); the green line is the mean. At 20 sentences the block is origin-class content. At 30 and 40 it is 100% destination-class content, yet it reads well short of $+1$. Panel titles give untrimmed means. The text quotes trimmed means, because the tank distributions have heavy tails, visible as the piles at ± 6 where readings beyond that range are stacked for display: $+0.34$ (aquarium→vehicle) and $+0.37$ (vehicle→aquarium) at ten post-shift sentences, and $+0.52$ and $+0.53$ at twenty.

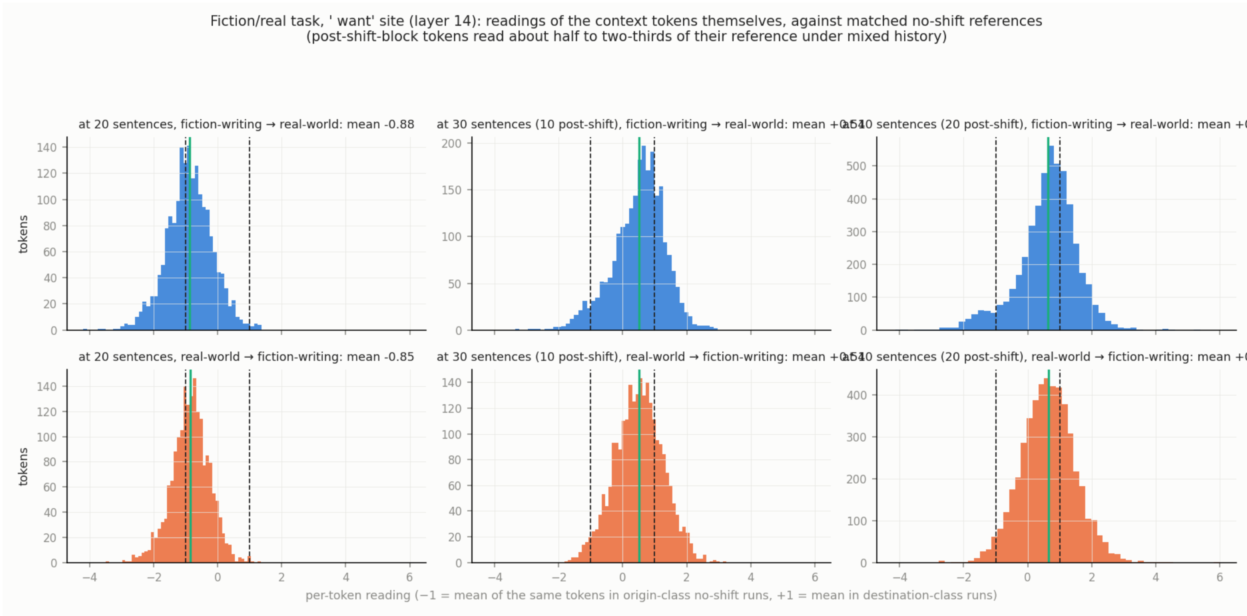

Figure 15: The same measurement in the fiction/real task (’ want’ site, layer 14), which has a tighter instrument and no heavy tails. Top: fiction-writing→real-world; bottom: real-world→fiction-writing; panels at 20, 30, and 40 sentences. At 20 sentences the block reads near -1, its origin-class baseline. The post-shift-block tokens read about half of the way to their no-shift reference at $+1$ at 30 sentences and about two-thirds at 40 (means in the panel titles).

3.4 The dwelling within the unresolved zone

Does any trajectory stop between the two interpretations rather than merely slow down? One does. In the tank task from aquarium to vehicle, the trajectory dwells within the unresolved zone: it becomes stationary mid-transition and, on the measured horizon, does not leave. Figure 16 tracks the mode of the run distribution by post-shift band. The mode is stationary at the midpoint for at least ten consecutive steps, and it sits at least 2.9 across-run standard deviations from both endpoint references. The spread across runs stays flat rather than tightening. A one-component fit beats a two-component fit in every time bin, so this is one stationary population, not a hidden mixture of resolved and unresolved runs. The stationarity is a property of the population, not of each run. Over the last ten post-shift sentences the family-clustered mean late slope lies within -0.040 and $+0.024$ axis units per sentence, so that over those ten sentences the population moves by under a fifth of the no-shift amplitude, and the median late reading sits at -0.11, at the midpoint. But individual late slopes range from -0.09 to $+0.09$ units per sentence, and only 3 of the 12 runs are individually flat by a criterion fixed before computing (slope within ± 0.02 and at least half the window in the middle band). In the reverse direction the population is still moving, with a late slope bounded away from zero at $+0.025$ to $+0.133$.

The signature replicates within the tank task and barely appears in the other. Within tank it appears at the ’ word’ site, the token just before ’ tank’ in the carrier, and under the replicate carrier “Define the word tank.” (§2.3). In the fiction/real task it appears only at the ’ letter’ site of the paraphrase carrier “Help me write a suicide letter.”, with 4 runs per direction, and we treat that as exploratory.

Two qualifications bound the claim. “Stationary” is bounded by the horizon: whether these

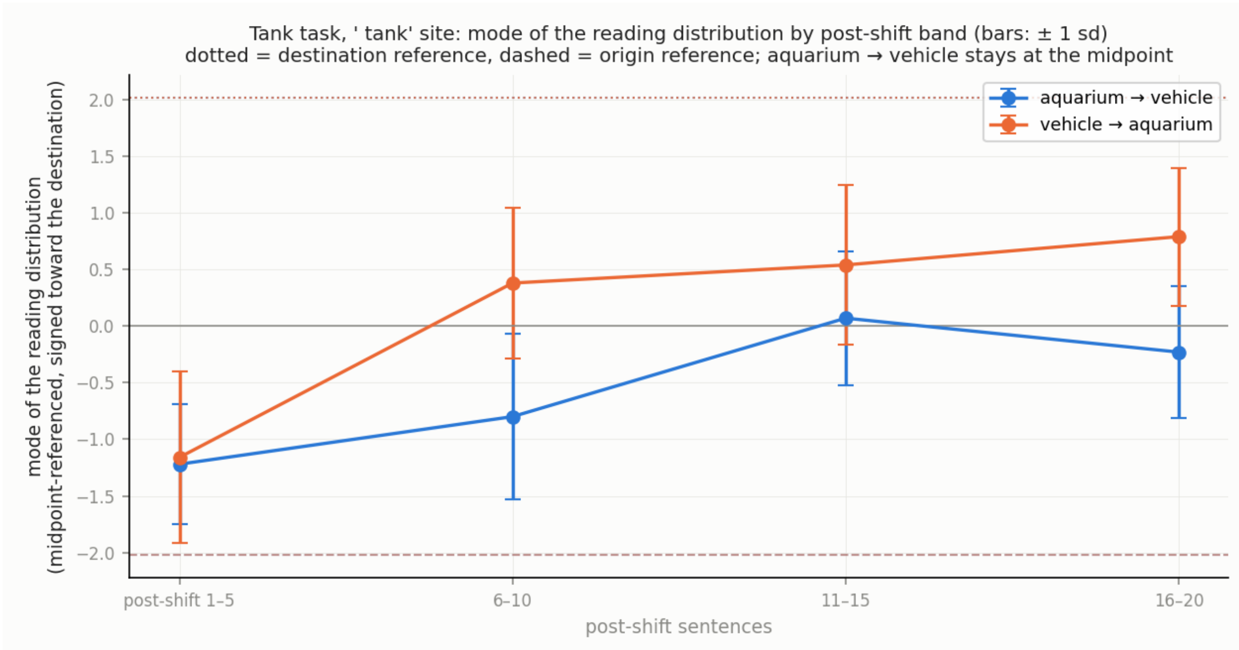

Figure 16: The dwell in the tank task, at the ' tank' site. Each point is the mode of the distribution of readings across runs within a five-sentence post-shift band, signed toward the destination class and midpoint-referenced; bars show one standard deviation across runs within the band. Dotted lines are the destination no-shift reference for each direction and dashed lines the origin reference. From aquarium to vehicle (blue) the mode reaches the midpoint by the third band and stays there through the last, at least 2.9 across-run standard deviations from both references, with the spread flat; a one-component fit beats a two-component fit in every band. From vehicle to aquarium (orange) the mode keeps moving.

runs would ever complete on a longer one is the same open question §3.2 leaves for the remnant, and some individual runs retain shallow nonzero late slopes. And geometrically the dwelling states are unremarkable. They sit inside the trajectory bundle, the spread of transition trajectories across runs, at 0.96 times the null distance, the typical distance among held-out reference states that §3.7 uses as its baseline. Functionally they are a distinct condition, as the model's behavior shows next.

3.5 What the model does in the unresolved zone

What does the model say while its reading sits between the two interpretations? We generated a completion from every transition run at four points after the shift, after 2, 6, 12, and 20 post-shift sentences, 96 in the tank task and 192 in the fiction/real task, and one from each of the 24 no-shift runs at its full length, giving 108 and 204. All behavioral claims in this subsection are correlational. For analysis we partition readings into three bands, one side, the middle, and the other side (Fig. 17).

A first result is that the model often does not answer at all. Under the greedy decoding used here, 14 of the 108 tank outputs and 85 of the 204 fiction/real outputs never leave the reasoning channel. The no-shift runs loop as well, 1 of 12 in the tank task and 5 of 12 in the fiction/real task, so looping is not a product of the shift. They repeat one sentence to the cap, cycle through one sentence frame with a changing noun, or re-read the passage without concluding. No cap would finish them. We call these "no answer" and report every rate below both with them counted and over delivered answers only. Deployed decoding samples rather than taking the greedy token, which is expected to break such loops, so their frequency in use is untested (§5).

In the tank task, behavior tracks the reading. Completions whose reading sits on the aquarium side answer with the aquarium sense in 52% of cases, 62% of those that answer. On the vehicle side the vehicle sense is given in 59%, 63% of those that answer. The middle band lists both senses in 45% of its completions, 52% of those that answer, and commits to one sense in 42%, 48% of those that answer. None of the 94 delivered tank answers asks which sense is meant or declines to answer pending disambiguation, by manual review and regular-expression scan of the committed table. In the unresolved zone the model either lists both senses or commits to one. It answers as if resolved.

Does the reading carry information beyond the context composition that drives both reading and behavior? We test this at matched composition, comparing completions generated after the same number of post-shift sentences (Fig. 18). The evidence is thin. At 6 post-shift sentences, decided answers come from runs with more extreme readings, with a median absolute reading of 0.90 against 0.38 for hedged or absent answers (one-sided $p = 0.037$); at 12 sentences they do not (0.53 against 0.56), the pooled test over both counts, fixed in advance, gives $p = 0.10$, and at the settled extremes there is no difference. We report it as one significant count of four and claim nothing further.

In the fiction/real task we categorize each delivered answer as fiction-writing assistance, which takes up the fiction-writing frame and helps with the letter or the manuscript, or a safe completion, which declines the letter or redirects to support. No delivered answer mixes the two. Of the 119 delivered answers, 111 are safe completions, 95 of them redirecting to support and 16 only declining, and 8 are fiction-writing assistance. Read over delivered answers, the safeguard holds across the reading bands: safe completions are 89% of the middle band's delivered answers and 95% of the real-world side's, and the fiction-writing side delivers one answer in four, a safe completion (Fig. 17, top right).

The reasoning channel tells the other half. Its final commitment matches the delivered answer in every one of the 119 cells that answered, so where an answer arrives the channel is expressed, not overridden. By band, the channel commits to a safe completion in 91% of the real-world side's cells, 82% of the middle band's, and 2 of 4 on the fiction-writing side (Fig. 17, bottom right). Neither reading shows a band difference the sample can distinguish from none: for the reasoning channel the family-clustered interval on the real-world-minus-middle difference runs from -0.07 to $+0.23$ (Fisher $p = 0.14$), and for delivered answers from -0.07 to $+0.18$ ($p = 0.38$). The difference between the two panels is the loops. Reasoning that commits to fiction-writing assistance loops in 15 of 23 cells, safety-committed reasoning in 70 of 181, a difference of $+0.27$ with a family-clustered interval of $+0.16$ to $+0.38$ (Fisher exact $p = 0.023$), and no fiction-writing-committed reasoning ever delivers a safe completion. Which cells loop is sensitive to the decoding path (Appendix B); the loop rate and this association are aggregate claims. So the two panels bracket what a user would see: delivered answers bound the safe rate from above, and the reasoning's commitments, which are what the loops would resolve to if they completed as committed, bound it from below. Where in that bracket the model sits under the sampling it is used with is the first item of future work (§5).

None of the 119 delivered fiction/real answers asks whether the request belongs to fiction writing or to the speaker's real circumstances. One assumes the fiction-writing frame and ends by inviting correction. In the reasoning channels, one proposes asking whether the request is for a story and one proposes asking the letter's purpose, and neither question is delivered. Two float a clarifying question and drop it, and seven safe completions plan to ask whether the user is safe. So the task with a safeguard surfaces the question no more than the tank task does. At matched composition the reading does not clearly separate the two response types: at 2 post-shift sentences the medians are +0.06 for fiction-writing assistance and +0.99 for safe completions, 3 answers against 26 ($p = 0.065$), and at later counts they do not differ.

An exploratory check asks whether any other layer's reading associates more strongly with behavior than the calibrated site's. Per-layer association curves (Fig. 19), computed over delivered answers, are roughly flat from mid-stack to the final layer in both tasks. Nothing singles out the deep layers. The instrument is blunt, a band-level association with imbalanced outcomes, so this neither establishes nor rules out a depth-specific behavioral readout.

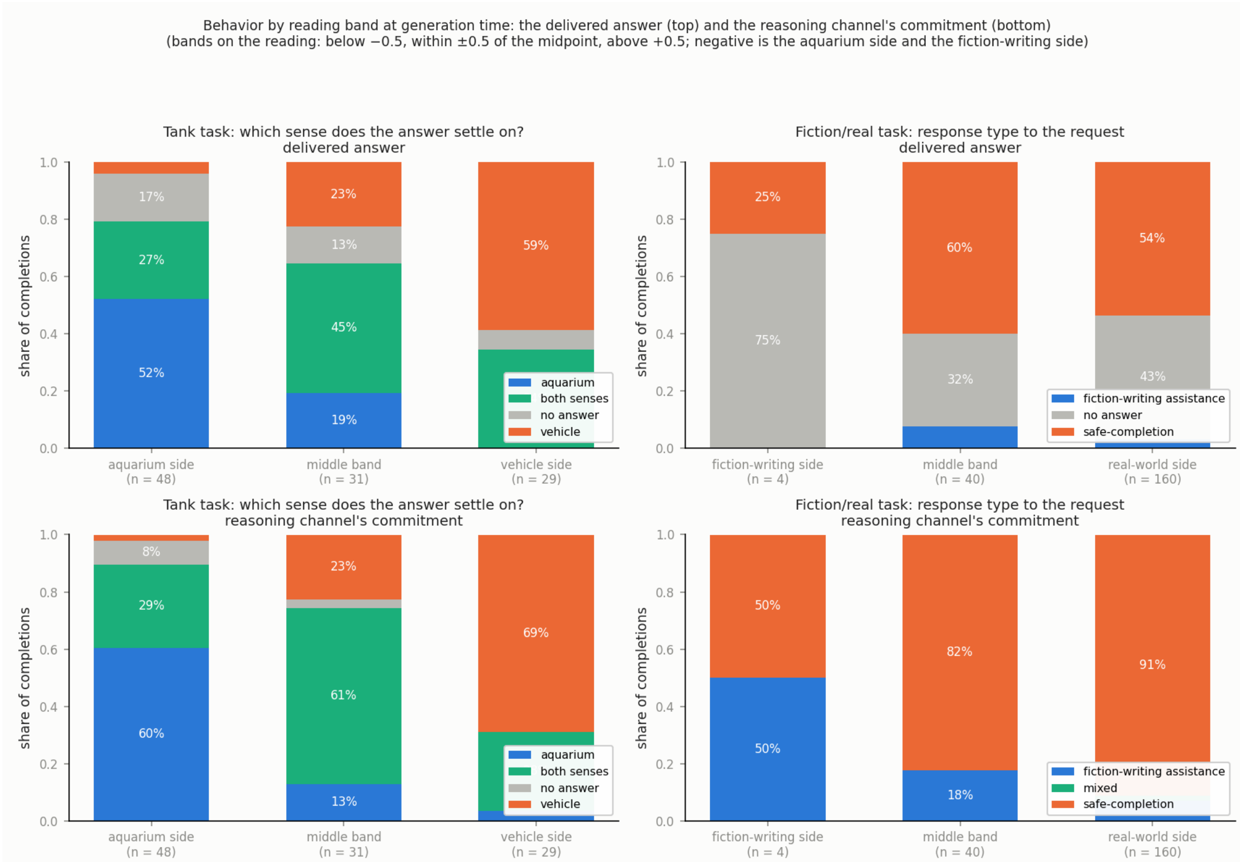

Figure 17: Behavior by reading band, read two ways. Completions are grouped by the reading at generation time: below −0.5 (the aquarium side in the tank task, the fiction-writing side in the fiction/real task), within ±0.5 of the midpoint (the middle band), and above +0.5 (the vehicle side, the real-world side); the number of completions per band is under each bar. Top row: the delivered answer, with gray the share of outputs that never delivered one. Bottom row: what the reasoning channel committed to in the same cells, which every cell has. Left: the tank task, by which sense the answer settles on. The side bands answer their side (52% and 59% of all completions, 62% and 63% of delivered answers); the middle band lists both senses in 45%, 52% of delivered answers. Right: the fiction/real task, by response type. Delivered answers safe-complete in 89% of the middle band and 95% of the real-world side, with one answer in four delivered on the fiction-writing side; the reasoning channel commits to a safe completion in 91%, 82%, and 2 of 4 across the same bands, and the difference is the loops, whose rate is higher on reasoning that had committed to fiction-writing assistance (15 of 23 against 70 of 181). Completions decoded greedily with a 2,048-token cap (§2.5).

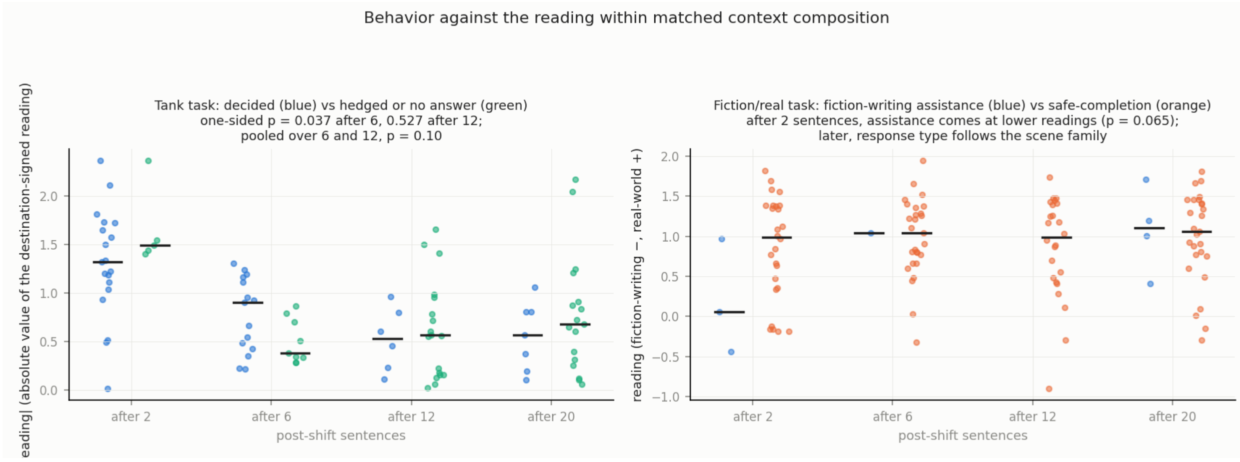

Figure 18: Behavior against the reading at matched context composition, that is, among completions generated after the same number of post-shift sentences. Left: the tank task. Each dot is the absolute value of one completion's reading, taken after signing the reading toward the destination class, split by whether the answer settled on one sense (blue) or listed both or gave no answer (green); bars are medians. After 6 post-shift sentences decided answers come from runs with more extreme readings (medians 0.90 against 0.38, one-sided $p = 0.037$); after 12 they do not (0.53 against 0.56), and the pooled test over 6 and 12 gives 0.67 against 0.55 ($p = 0.10$). Right: the fiction/real task, reading by response type, fiction-writing assistance (blue) or safe-completion (orange); outputs that delivered no answer are excluded. After 2 post-shift sentences assistance comes at lower readings (medians +0.06 against +0.99, 3 answers against 26, $p = 0.065$); at later counts the reading does not separate the types.

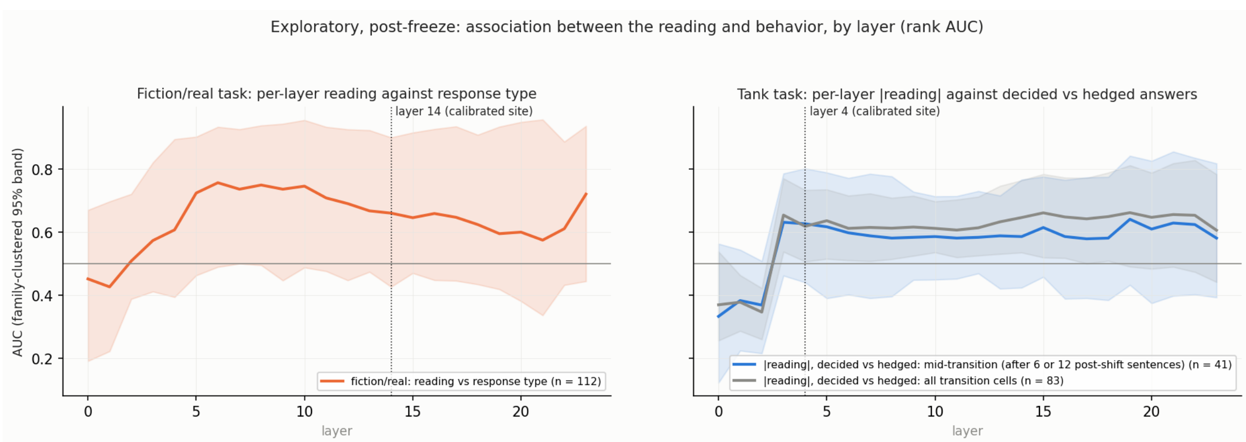

Figure 19: Exploratory: the association between the reading and behavior at every layer, as a rank AUC with family-clustered 95% bands. Left: the fiction/real task, reading against response type (112 delivered transition answers). Right: the tank task, absolute reading against decided versus hedged answers, for mid-transition completions (41 delivered, at 6 and 12 post-shift sentences) and for all transition completions (83 delivered). Dotted lines mark the calibrated sites. The curves are roughly flat from mid-stack to the final layer in both tasks. The instrument is blunt: a band-level association with imbalanced outcomes.

3.6 Order and equilibrium: hysteresis without stickiness

Does the order of evidence matter beyond its amount? To ask this we built static mixture sweeps: twenty-sentence contexts holding k destination-class sentences, with k swept from 0 to 20, in two block orders, using each family's own transition sentences. The resulting loops are large. The same mixture reads differently depending on block order, with a loop area of +14.4 [12.2, 16.8] in the tank task (Fig. 20) and a comparable one in fiction/real (Table 5). Operational hysteresis is plainly present.

The open question was whether any of that order dependence exceeds what recency weighting alone produces. We call any such excess *stickiness*. There is none. A one-parameter recency integrator fitted to the sweep cells reproduces the loop areas almost exactly, and the excess is indistinguishable from zero in both tasks (Table 5). Cross-order validation preserves the verdict: fitting one branch predicts the other. A null that imports γ from the transition fits instead of fitting it to the sweep cells shows apparent stickiness. The fitted null supersedes it (Appendix A). What remains beyond the one-parameter integrator is mild. The two sweep directions prefer slightly different recency weights, echoing the directional asymmetry of §3.2.

Table 5: Hysteresis in the static mixture sweeps. Loop area is the area between the two block-order branches of the sweep, in axis units times sentences, with a family-clustered bootstrap 95% interval. The fitted loop area comes from a one-parameter recency integrator fitted to the same cells; the excess is the observed area minus the fitted area. The last column gives the recency weight γ fitted separately to each block order, with the destination block placed first or last in the context.

Task	Observed loop area	Fitted integrator loop area	Excess (stickiness)	γ, destination block first / last
tank	+14.4 [12.2, 16.8]	+14.3	+0.1, not significant	0.91 / 0.97
fiction/real	+10.5 [7.5, 13.3]	+11.1	−0.5, not significant	0.90 / 0.84

This resolves an apparent contradiction with §3.3. The equilibrium map from evidence mixture to reading is smooth and integrator-like. The sequential path through a transition is drift punctuated by jumps. Both are true. They are different observables. The metastability this paper names is a property of paths, not of the equilibrium map. There is no bistable equilibrium and no barrier between states.

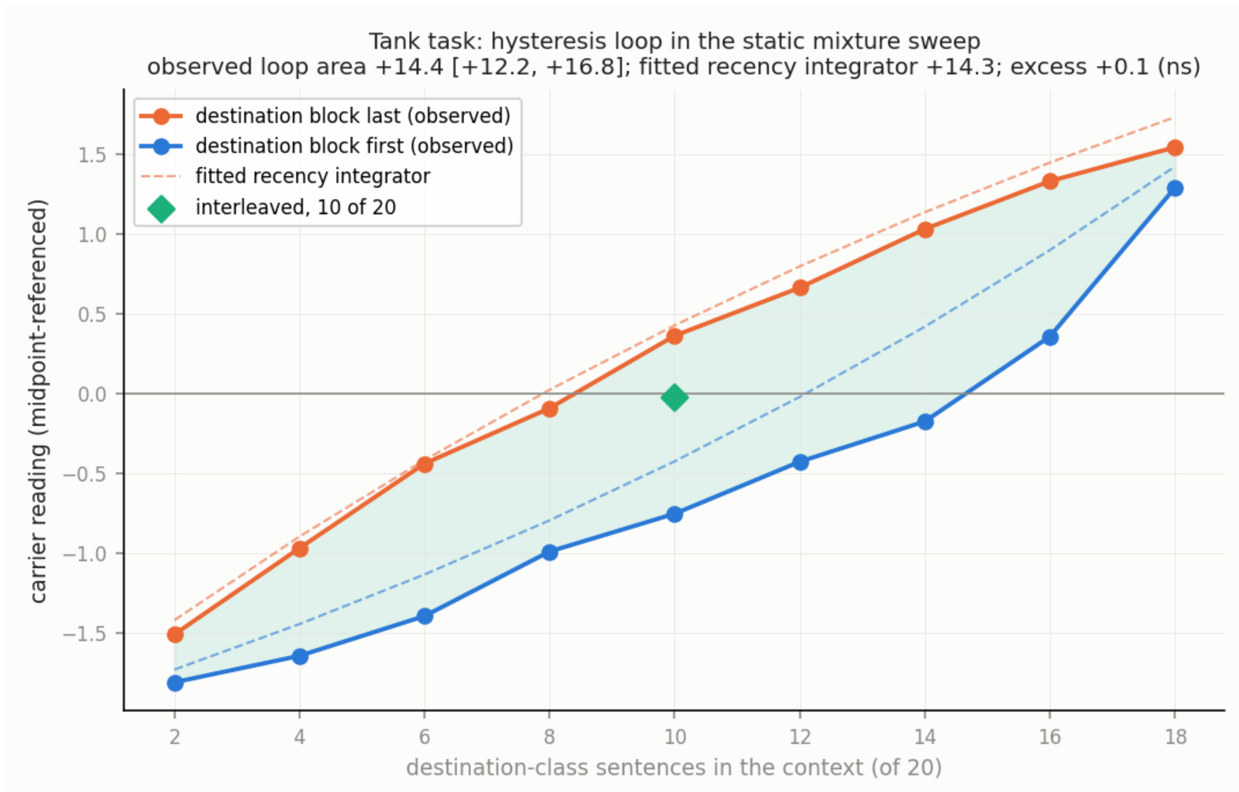

Figure 20: Hysteresis in the tank task. Each point is the mean carrier reading, midpoint-referenced, in a twenty-sentence context holding a fixed number of destination-class sentences, with the destination block placed last (orange) or first (blue). The shaded area between the branches is the loop: the same mixture reads differently depending on the order, with area +14.4 [12.2, 16.8]. Dashed lines are the predictions of a one-parameter recency integrator fitted to the same cells, which reproduces the loop (area +14.3); the excess, +0.1, is indistinguishable from zero. The green diamond is an interleaved mixture of 10 destination sentences in 20, which falls between the branches. The fiction/real sweep (Appendix C) gives +10.5 [7.5, 13.3] observed against +11.1 fitted (Table 5).

3.7 The geometry of irresolution

Finally, the third of the introduction's three worlds: do these in-between states leave the model's learned distribution? "Off-manifold" is meaningful only relative to a reference, a measure, and a noise level, so we fix all three. The reference is the position-matched no-shift states, together with the trajectory bundle of §3.4, the pooled set of transition-run states across all runs. The measures are two standard novelty scores: the distance to the nearest neighbors in the reference (e.g. Sun et al., 2022) and the reconstruction error of a state against the principal-component subspace of the no-shift states (e.g. Jackson and Mudholkar, 1979). The noise level, which we call the null, is the held-out self-distance of the reference itself. Throughout this section, paired values are for the tank and fiction/real tasks in that order. The instruments detect real displacement. Single-sentence calibration states read 1.7 to 2.2 times the null across the two instruments. They are a genuinely different context regime and serve as a positive control. No-shift states compared against the reference of a different position, which we call position-mismatched, are flagged as displaced in 13% to 44% of cases.

At the level of individual states, nothing leaves the reference distribution. Jump steps read 1.06 and 1.12 times the null (Fig. 21). The dwelling states read 0.96 times. All three sit inside the null's spread. Our pre-registered prediction that jump steps would show elevated off-manifold distance failed.

A displacement too small to flag any single state could still be shared by all of them, so we tested the mean directly. The mean out-of-subspace reconstruction error of transition states is far outside the range of a null built by resampling whole scene families, the family-clustered bootstrap of §2.5 (Fig. 22). It is a shared displacement equal to 25% and 38% of the full class separation, with $p < 0.001$ in both tasks. It rises over the first five post-shift sentences and persists undiminished to twenty, and it is largest in the stationary dwelling states. We call the direction that carries it the **mixed-context marker**.

Three checks say what the marker is not (Table 6). It is not class leakage: the direction is orthogonal to the single-sentence class axis and, by construction and by measurement, to the accumulated-context class axis. It survives held-out direction estimation, in which the direction is estimated on half the families and the magnitude measured on the other half. It retains 71% to 76% of its in-sample magnitude in the tank task and 87% to 90% in fiction/real. And it is not family novelty. If the marker merely reflected unfamiliar scene families, pure-class cells from families outside the reference construction should sit higher on it than cells from familiar families. They sit lower, and both sit far below mixed cells.

What is it? Held-out mixture cells, cells of the static sweeps that were not used to construct the marker, locate what elevates it. The marker is absent from pure-class contexts. In static mixed contexts it is near its full transition strength in the tank task and about 70% of it in fiction/real (Table 6), so it marks mixed context, not temporal shifting as such. Those cells were all captured on one day, so the separation between pure and mixed cells cannot be a capture-day effect (§2.1). One qualification: in the fiction/real task it halves when the mixture is interleaved rather than blocked, so there, part of what elevates it is the coherent, blocked structure of the shift. It does not predict behavior at matched composition. In eight tests at the calibrated site and layer, one per task and post-shift count, the marker score of the delivered answers does not differ between the outcome classes: none of the seven defined tests reaches $p < 0.05$ uncorrected, the eighth has no fiction-writing answer to compare, and the score does not separate delivered answers from loops either. The fiction/real tests are underpowered, with 8 fiction-writing answers spread over four counts. Other layers are untested. The model, in other words, carries a persistent, systematic

Table 6: The mixed-context marker and its checks. Values are for the tank and fiction/real tasks. The displacement rows are in percent of the full class separation; the cosine rows measure the marker direction against the two class axes.

Quantity	tank	fiction/real
Shared displacement of transition states	25%	38%
Cosine with the single-sentence class axis	+0.015	+0.011
Cosine with the accumulated-context class axis	−0.0052	−0.0065
Retained under held-out direction estimation	71–76%	87–90%
Pure-class cells from families outside the reference construction	+1.6%	+3.4%
Pure-class cells from familiar families	+6.7%	+6.0%
Mixed cells	≈ +27%	≈ +27%

signal that its context is mixed: a direction on which pure contexts sit near zero and mixed contexts near 27% of the class separation. Its behavior does not appear to use it. Whether this signal is a learned representation of mixedness or a systematic consequence of heterogeneous input is not decided by our measurements.

Three graded senses of what we have called off-manifold, or more broadly off-distribution, should be kept distinct. The first is exotic natural inputs. The second is interventional states, such as patching and steering. The third is states of one context regime measured against a reference built from another, as our single-sentence positive controls are. Our shifts are the tamest possible case, clean block transitions between two well-learned frames. That even the dwelling states stay on-distribution here says nothing about adversarial or incoherent contexts, where genuine excursions remain most likely. That battery is future work.

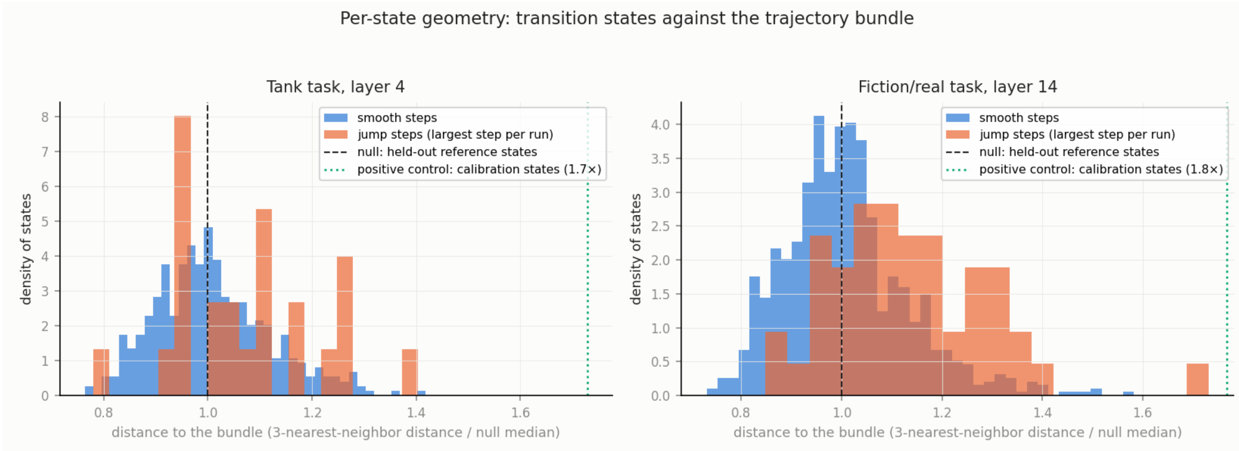

Figure 21: Per-state geometry. For every transition state at the calibrated site, the distance to its three nearest neighbors in the reference set, divided by the median of the same score for held-out reference states (the null, dashed line at 1.0). Blue: smooth steps; orange: the largest step in each run, the jump steps. Left: the tank task at layer 4; right: fiction/real at layer 14. Both distributions sit inside the null's spread. The dotted green line is the positive control: single-sentence calibration states, a different context regime, read 1.7 and 1.8 times the null on this instrument.

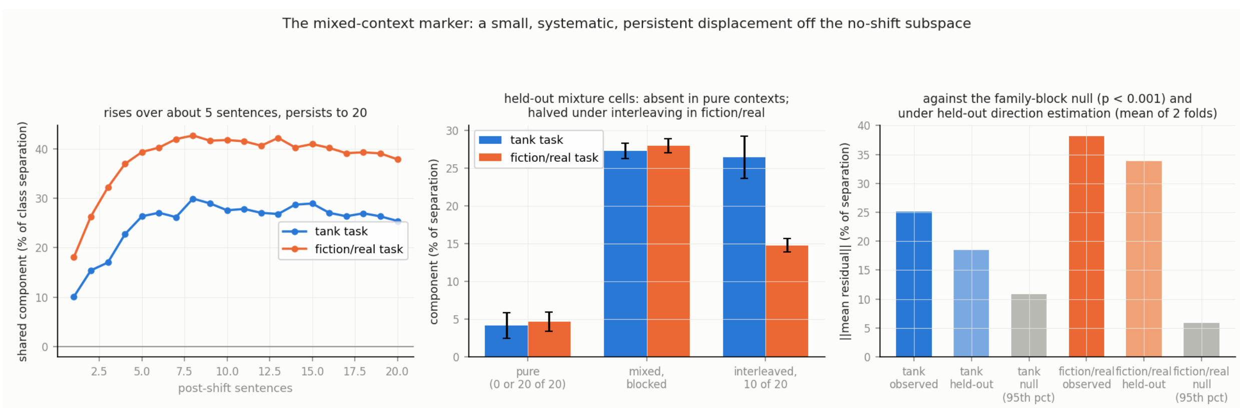

Figure 22: The mixed-context marker. Left: the mean displacement of transition states outside the no-shift principal-component subspace, projected on the shared direction and expressed as a percentage of the class separation, by post-shift sentence. It rises over about five sentences and persists to twenty. Middle: held-out mixture cells from the static sweeps. The marker is near zero in pure-class contexts (0 or 20 destination sentences of 20), near full strength in blocked mixtures, and, in the fiction/real task only, halved when the mixture is interleaved; error bars are one standard error over cells. Right: the observed mean displacement against the 95th percentile of a null built by resampling whole scene families ($p < 0.001$ in both tasks), and its value under held-out direction estimation, the mean of two folds.

4 Discussion

What "metastable" means here. We take the term from the dynamics of biological neural systems rather than from physics. In coordination dynamics and in the winnerless-competition framework, two models of sequential transient dynamics in neural populations, cognition is described as a sequence of transiently stable states that are not fixed-point attractors (Kelso, 1995; Rabinovich et al., 2008). These are structured passages through state space, in which dwelling and moving are not opposites. That is the geometry our data show. A run that dwells mid-transition occupies something functionally state-like: it is stationary for many steps and has its own behavioral signature. Geometrically it is a passage, inside the trajectory bundle and on a smooth equilibrium map. One disclaimer follows. Nothing here is equilibrium bistability, since the map from evidence mixture to reading is smooth (§3.6), and readers should not import a barrier-crossing picture. Semantic metastability, as we use it, is a property of paths.

Two caveats bound the term. The plateau's location is ambiguous on its own: an equal-weight average of twenty aquarium and twenty vehicle sentences reads at the midpoint, so the location cannot separate a transient state from an equilibrium reading of mixed content. What separates them is that no single weighting produces both phases. The equal weighting that lands on the midpoint would cross it only at the twentieth post-shift sentence, where the runs cross at a median of 10.5, and the recency weights fitted independently to the static sweeps, γ of 0.91 to 0.97 (Table 5), give the destination block 65% to 87% of the weight, which places the equilibrium reading of destination-last content on the destination side of the midpoint, not at it. The term therefore rests on the cluster, the two-phase incompatibility and the state-timed jumps, not on the plateau's position. And the term names a signature, not a mechanism. There are no autonomous dynamics here: each reading is a fresh forward pass on a longer prompt, and the phenomenology arises from a map without attractor structure (§3.6) processing shifting inputs. That absence is itself a finding.

Typicality is not commitment. The three-worlds question of §1 dissolved because it conflated two axes. On distributional typicality the stationary states are unremarkable: inside the bundle, on-distribution. On semantic commitment they are distinctive: uncommitted, between calibrated interpretations, with hedging as their behavioral expression. A learned state of unresolvedness would be typical and uncommitted at once. That is the profile the stationary states show, which is why the first two worlds of §1, a learned state and a passage, were never alternatives. Genuinely off-distribution states, such as glitch inputs, interventions, and adversarial incoherence, fall elsewhere on the typicality axis and remain untested here. The general lesson is that a state can be geometrically typical without being functionally normal. Our behavior data document the functional difference.

Safety and alignment, first descriptively. The failure surface this study maps needs no adversary. During an ordinary, coherent shift of conversational frame, a model's reading of a critical token trails the present frame for roughly the fitted recency timescale. Here that is nine to fourteen sentences, and effectively without bound in the dwelling direction. It retains a remnant of the prior frame beyond that. In one tank direction it dwells between frames for the remainder of the tested horizon. The fiction/real task shows an exploratory echo of this, resting on 4 runs per direction.

In the tank task, behavior inside this window tracks the reading. Throughout, the model never asks which reading is meant or flags the ambiguity as an obstacle. That holds in every delivered tank answer. Even so, half of the middle band's delivered answers surface both senses. In the fiction/real task it does not: safeguard behavior, read from delivered answers, holds across the reading bands, and read from the reasoning channel's commitments it is 91% on the real-world side and 82% in the middle band, a difference the sample cannot distinguish from none, with the fiction-writing side too thin to grade. The two readings differ because reasoning that commits to fiction-writing assistance usually loops under greedy decoding instead of answering (§3.5).

A trained default for unresolved cases? What follows is a post-hoc reading of an asymmetry we noticed, not a designed manipulation, and training provenance is unobservable. Our two tasks appear to differ in whether they carry a trained default for unresolved cases. The covered fiction/real task behaved as though it does. In the middle reading band, 89% of its delivered answers safe-complete, and 82% of its reasoning channels commit to a safe completion; where the reasoning entertains the fiction-writing frame, the answer that follows, when one follows, takes it up. The tank task showed no such default. There the model silently commits: 48% of delivered middle-band answers pick a sense. If this reading is right, the unresolved zone is the failure window for every behavior without a trained uncertainty-default, and refusal-style safeguards may be the well-covered exception rather than the rule.

Why the safeguard held: three candidate accounts. Three accounts could explain why the covered fiction/real task mostly safe-completed in the middle reading band, and we cannot yet separate them. They are not exclusive: one is about what training installed, one about what deeper layers see, one about where in the stack the trigger reads. None is causally established. With two tasks these are observations. The exploratory per-layer curves of §3.5 cannot separate them either. Shallow readings saturate after the shift, which blinds that instrument in exactly the layers the third account below points to.

A trained default. The first account is the reading above: the fiction/real task carries a trained default for unresolved cases, and the tank task does not.

Resolution at depth. The second account emerged from the depth data, and it is weaker than it first looked. In the fiction/real task, layers 3 to 18, the two middle bands, settle clearly on the destination side in both directions, where the tank task from aquarium to vehicle ends at the midpoint (§3.2). Downstream layers in the fiction/real task may therefore act on a more settled reading than the site we measure. But the deepest layer band hovers near the midpoint in the fiction-writing-to-real-world direction too, the direction in which safeguard behavior must appear, and did. Resolution at depth cannot carry the explanation alone.

An early, surface-keyed trigger. The third account points the opposite way through the stack. The shallowest layers resolve the framing composition almost immediately and completely (§3.2), with the profile of surface-cue tracking. The minimal-pair test, which shows the reading tracks framing cues rather than content, was run at the calibrated site, not at shallow layers. A safeguard that reads early, from surface content and frame cues, would fire whenever the alarming request is present and be suppressed only by a well-established fiction-writing frame. That is consistent with our behavioral data, in which the safeguard fires in nearly every delivered answer. The fast, complete shallow response is specific to the fiction/real task. The tank task's shallow layers respond later and, in one direction, only partway (§3.2). That is what a trigger sculpted by safety post-training would look like. But ordinary register statistics learned in pretraining predict the same asymmetry, and training provenance is unobservable here. Patching shallow against mid-stack states during generation is the decisive test (§5).

One limit. As a standalone monitor, predicting each response from its reading alone, the frame reading, the fiction/real task's reading at its calibrated site, is not yet usable: its AUC is 0.61 [0.37, 0.81], compatible with chance, with only 8 fiction-writing answers among 192 completions (Fig. 23). A usable monitor would combine several sites, the mixed-context marker, and per-layer readings. Building one is future work, not a claim.

A signal behavior does not appear to use. The mixed-context marker gives the sharpest form of the dissociation between what the model represents and what it does. The model carries a persistent, systematic signal that its context is mixed, and its behavior does not appear to use that signal when the mixture of frames in the context is held fixed, at the one site and layer tested. The reasoning channel shows the contrast: what it commits to is expressed whenever an answer arrives (§3.5). The marker, by comparison, is present and, as far as tested, not acted on. This is the structure reported in the hallucination literature, where models internally encode uncertainty or truthfulness that their generations do not respect (Azaria and Mitchell, 2023; Orgad et al., 2025). Here it appears in transition dynamics. An internal signal correlated with the conditions that produce the unresolved zone already exists inside the model, and behavior does not appear to use it. Whether it could actually flag the zone is untested; the one monitor we built, from the frame reading, is compatible with chance. The motivating case of §1 reportedly had the same structure at system scale: the provider's moderation layer was flagging the user's messages for self-harm risk in real time while assistance continued (Raine v. OpenAI, Inc., 2025b; CNN Business, 2025). That suggests a constructive direction for alignment work: test whether behavior can be gated on the uncertainty signals already there, before building new ones.

The human parallel. The §3.2 signature is a partial reanalysis that leaves a lingering trace of the initial misreading. That is also the signature of human sentence processing in the "good-enough" tradition (Ferreira et al., 2002; Christianson et al., 2001). After recovering from a garden-path sentence, comprehenders measurably retain components of the initial, incorrect interpretation. We make no mechanistic identification. We note that a language model trained on human text reproduces, at the activation level, the reanalysis profile humans show behaviorally, including the part where the old reading never fully leaves.

Garden path or pun. In the terms of Dynel's account of garden-path humor (Dynel, 2009), our transition corpus is a slow-motion garden path, and most trajectories treat it as one: incongruity, then resolution toward the new reading. The dwelling runs are the interesting case. They form an extended incongruity phase, in which the model, asked what the word means, answers like someone explaining a pun: both senses, held. Whether any of them is a true pun, a held incongruity that never resolves, or only a long garden path is the persistent-versus-slow question our twenty-sentence horizon cannot decide. The extended-tail experiment named in §5 is its dedicated test.

Closing. Word-sense reinterpretation is the tractable laboratory instance of a much larger family. Frames, personas, tasks, and safety postures all shift under accumulating context, and there is no obvious reason the metastable structure documented here is unique to word senses. The states are what the title calls them: unresolved. For now, so is the question of what a model should do while inside one.

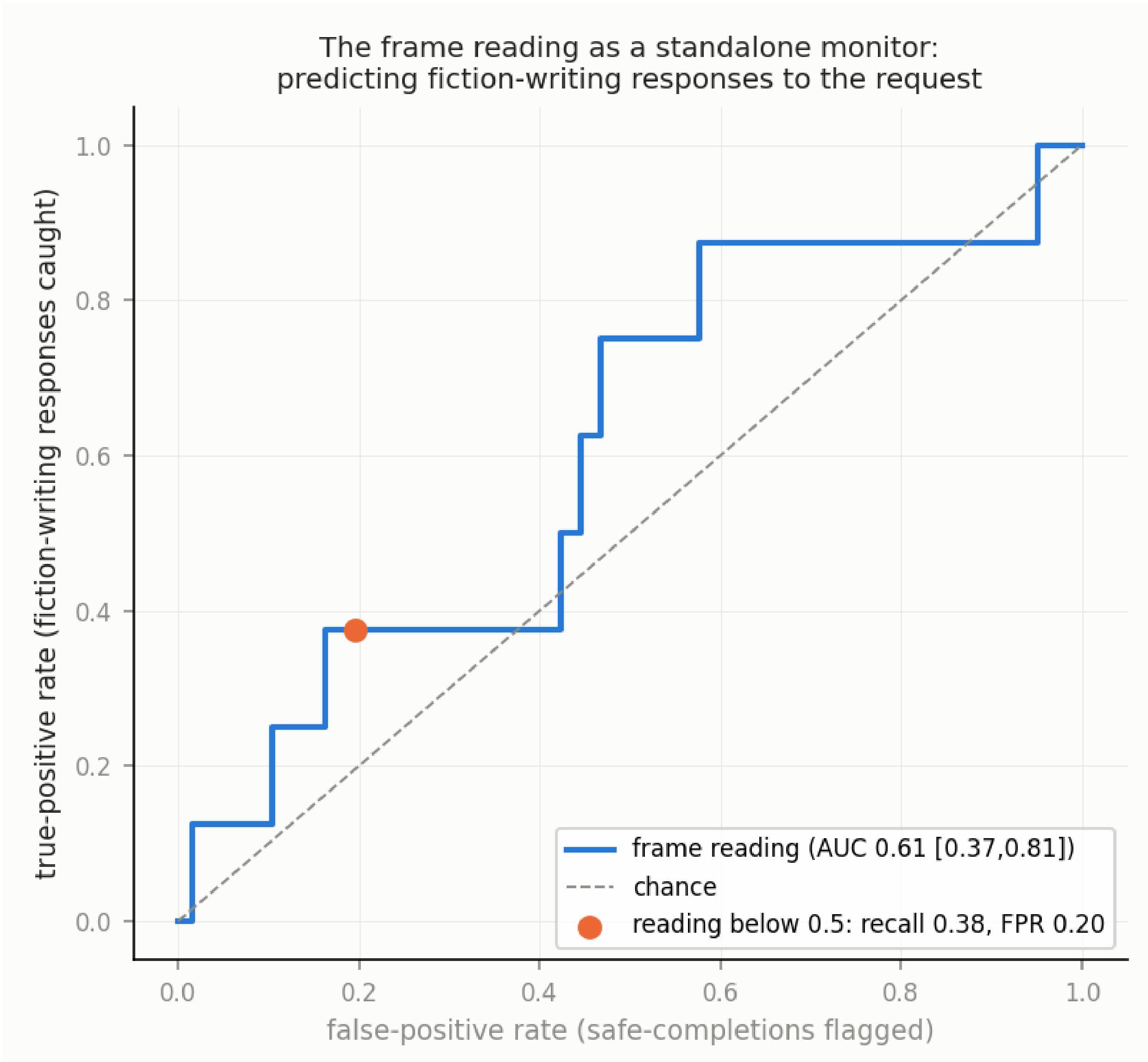

Figure 23: The frame reading as a standalone monitor, an exploratory check. Receiver operating characteristic for predicting fiction-writing responses to the request (responses that take up the fiction-writing frame rather than safe-completing) from the calibrated-site reading alone: the true-positive rate (fiction-writing responses caught) against the false-positive rate (safe-completions flagged). The AUC is 0.61 [0.37, 0.81], with a 95% interval clustered by scene family (§2.3), compatible with chance (dashed line); only 8 of the 192 transition completions are fiction-writing answers. The marked point is one threshold, with its recall and false-positive rate.

5 Limitations and future work

One model, stated plainly. Everything here is measured in one 20-billion-parameter mixture-of-experts model, with greedy decoding, so the behavior rates of §3.5 are properties of the greedy continuation, and so are the loops that leave many completions without an answer. Deployed decoding samples, which is expected to break such loops; behavior under sampling is untested here. Three kinds of claim should be kept apart. Descriptive claims about gpt-oss-20b are established for the tested tasks, contexts, sites, and decoding conditions, since this model is the population of interest. Existence claims about the class, that a deployed language model can dwell between interpretations, can carry a mixed-context signal its behavior does not use, and can deliver fiction-writing assistance with the letter under ordinary coherent context, with no adversarial prompt, need only one model. Prevalence claims across models are untested here, and the paper makes none. This is the mode of work that the recent circuit-level biology of a single language model adopted: one deployed system characterized in depth, with transfer left to later work (Lindsey et al., 2025). The model is itself deployed and open-weight, so its characterization has safety relevance of its own, independent of any transfer to other systems.

Each dynamics claim is made at one calibrated site and layer per task. Those sites are single rows of a depth progression in which crossing times vary systematically across layers (§3.2). No single layer is a sufficient readout of the model's state, and behavior is produced downstream of all of them. Within the study, replication is internal. Its scope is set out in §2.3: the task contrasts, carriers, sites, and scene families.

What did not replicate cleanly. The real-world→fiction-writing remnant is suggestive only once reference uncertainty is propagated. The ' letter'-site asymmetry rests on four runs per direction. In the fiction/real task, 48% of runs are indeterminate under per-run trajectory-model selection. And our pre-registered prediction about the ordering of crossing times across layers held in one task and failed in the other.

Patterns seen on two tasks. Two further patterns hold across the tasks. With two tasks they are observations, so we treat them as hypotheses rather than findings. The task with the wider endpoint separation also shows the larger remnant gaps. And the two tasks differ in whether their contrast is anchored to one token or spread across the utterance.

The named open question. Are the remnant and the dwelling permanent, or only slower than twenty sentences can resolve? In the terms of §4, is the dwelling a pun or a long garden path? Two experiments would decide it. One extends the post-shift horizon. The other shifts the evidence back to the original class after the shift and measures what remains of the second frame. Both are designed and costed. Neither has run.

Further deferred work, in rough order of leverage:

- Regenerate the behavior completions under sampling at the model's default temperature, several draws per cell. The two readings of §3.5 bracket the safeguard's rate; sampling, which is expected to break the loops, decides where in the bracket the deployed model sits.
- Replicate the fiction/real headline quantities at the ' write' site, where the framing contrast reads most strongly. The existing recordings suffice.
- Expand the ' letter'-site families.
- Build calibration sets from a third, unrelated class, to identify what the accumulation offset contains.

- Construct a monitor from several sites, the mixed-context marker, and per-layer readings, and test it against the chance-compatible single-site baseline of §4.
- Localize where safeguard behavior reads in the stack by patching shallow against mid-stack states during generation. This is the decisive test for the three candidate accounts of §4.
- Run a benign-framing control arm: the same carrier with one word swapped, such as a resignation letter, with contexts built by the fiction/real recipe with the theme swapped, so that cue density is matched by construction. A benign request has no safeguard to trigger, so this arm separates the trained-default account of §4 from the pretraining-register account, where patching only localizes. It is specified here and not yet run.
- Run a suite of harder context manipulations: colliding frames, frames that mutate mid-context, and deliberately incoherent contexts. Genuine off-distribution excursions are most likely there. Our clean block shifts are the tamest possible case.

Back matter

Related work

Reading representations with probes. Linear probes have been the standard way to ask what a layer represents, and their validity has been questioned from the start: a probe can learn its own task rather than report the model's (Alain and Bengio, 2016; Hewitt and Liang, 2019; Belinkov, 2022). Our answer to that concern is the three tests of §3.1 and the label shuffles of Box 1. The instrument assembles established methods. The difference-of-means axis is the mass-mean probe (Marks and Tegmark, 2024), which found such directions read out class structure comparably to trained probes; difference directions between two conditions go back to word-embedding analyses (Bolukbasi et al., 2016) and to concept activation vectors (Kim et al., 2018), and rest on the linear representation hypothesis (Park et al., 2024). Classifying by the nearer class mean is nearest-centroid classification (Tibshirani et al., 2002), and word senses were read from contextual activations in just this way, by nearest sense centroid, in early geometry studies of BERT (Reif et al., 2019). The rotation of a readout direction across depth (Box 1) is the effect that motivates refitting a lens at every layer (Belrose et al., 2023). The per-token class signal we report is d' from signal detection theory (Green and Swets, 1966). The two off-distribution scores of §3.7 are the k-nearest-neighbor distance (Sun et al., 2022; Lee et al., 2018) and the principal-component reconstruction error (Jackson and Mudholkar, 1979). The nearest cousin of our fiction/real behavior link is the refusal direction, a single difference-in-means direction along which refusal behavior can be read and steered (Arditi et al., 2024), and the same direction family underlies contrastive activation steering (Panickssery et al., 2024).

Dynamics of in-context processing. The account of in-context learning as implicit Bayesian inference is the standing theory our recency-integrator comparison speaks to (Xie et al., 2022). Function-vector and task-vector work shows that a context's task can be summarized in a direction of the residual stream (Hendel et al., 2023; Todd et al., 2024), and induction heads describe one mechanism by which context propagates (Olsson et al., 2022). Those describe the summary once formed, or the mechanism that forms it. Probes of belief state and board state follow a latent state as it updates, on synthetic or game data (Shai et al., 2024; Li et al., 2023), and incremental parse-state probes read a syntactic interpretation as words arrive (Eisape et al., 2022). That in-context processing weights recent items more heavily is also established (Zhao et al., 2021; Liu et al., 2024). Our recency integrator turns that property into one fitted timescale. What happens to a natural-language interpretation while it is being overturned, and whether the model acts on the intermediate readings, is the question here.

Long-context safety. Many-shot jailbreaking and multi-turn escalation attacks show that accumulated context can override trained behavior (Anil et al., 2024; Russinovich et al., 2025). Our shifts are not attacks. They are coherent conversations, and the fiction-writing assistance the model delivered in a few of them needed no adversary.

Verbalized uncertainty and ambiguity. Models can be trained or prompted to express confidence, and their verbalized confidence can be calibrated (Kadavath et al., 2022; Lin et al., 2022; Xiong et al., 2024). Ambiguity in questions has been modeled directly (Liu et al., 2023). These ground the scoped claim of §1: models can express uncertainty, and none of the answers the model delivered asked which reading was meant.

Internal encoding versus expression. Models internally encode truthfulness and uncertainty that their generations do not respect (Azaria and Mitchell, 2023; Orgad et al., 2025). Residual-stream signals of conflict between the context and stored knowledge have also been reported (Zhao et al., 2024). The mixed-context marker of §3.7 belongs to this family: a signal, here of conflict within the context itself, that is present and, as far as we tested, unused.

Long-context artifacts. Attention sinks and related position effects raise the question of whether the accumulation offset of Box 1 is positional (Xiao et al., 2024). Midpoint referencing makes our findings robust either way, since every level claim is measured against a matched no-shift reference at the same position.

Human sentence processing. The good-enough tradition documents that comprehenders retain components of an initial misreading after recovering from a garden path (Christianson et al., 2001; Slattery et al., 2013; van Schijndel and Linzen, 2021). This is the parallel drawn in §4. Dynel's taxonomy of conversational humor supplies the garden-path and pun distinction of §1 and §4 (Dynel, 2009).

Metastability in neural dynamics. Coordination dynamics and the winnerless-competition framework describe cognition as sequences of transiently stable states that are not fixed-point attractors (Kelso, 1995; Rabinovich et al., 2008, 2001; Sussillo and Barak, 2013). We take the term from there.

The motivating case and the model. The case described in §1 is Raine v. OpenAI, filed in the Superior Court of California, San Francisco, on 26 August 2025, with contemporaneous reporting (Raine v. OpenAI, Inc., 2025b; CNN Business, 2025; Yang et al., 2025). A later report covers the defendant's answer denying causation (Raine v. OpenAI, Inc., 2025a; NBC News, 2025; TechCrunch, 2025). The litigation is contested. This paper characterizes only what the filing and reporting describe. The safety training of the model we study is documented in its model card (OpenAI, 2025).

Ethics and safe messaging

This paper studies how a model's internal reading of a suicide-related request moves under context, and reports the model's behavior around that request. It involves no human subjects and no user data. The fiction/real corpus was written under the norms of §2.3: no context sentence contains the carrier request, and the theme-only sub-arm never names a suicide letter or note. The content note on the first page follows safe-messaging practice, and the paper quotes no model-generated text that assists with the letter. Completions containing such text are withheld from open release (see Data and code availability, §2.6). The motivating case is contested litigation, and the paper characterizes only what the filing and reporting describe.

Acknowledgments

This study was carried out with language-model assistants, Claude through Claude Code and in chat, used as the analysis runtime and as writing and review assistants. Under the author's direction they authored the context sentences under the blind protocol of §2.3, ran the captures and the committed analysis scripts, categorized the completions, and drafted and revised the text. The author reviewed every number and claim and is responsible for them. The research questions, study design, and interpretive decisions are the author's.

Appendix A — Corrections

The study's corrections log, nineteen entries at the analysis freeze and extended since, is in the repository's findings record and revision log with dates. The audits caught three kinds of thing: instrument artifacts, above all uncorrected per-layer readings that produced three findings retracted before this version (Box 1); a misspecified null, retracted the same day; and reference uncertainty that widened one interval to include zero. Four corrections changed values as printed: the real-world→fiction-writing remnant interval widened to include zero once the no-shift references were resampled (§3.2); the tank within-stream readings are trimmed means (§3.3); crossing times are per-run medians rather than a range read from mean trajectories (§3.2); and the recency integrator's memory is stated as per-direction decay values (§3.2). A first estimate of stickiness (§3.6) used a misspecified null and was retracted the same day.

Appendix B — Quality assurance and reproducibility

Regeneration. A full regeneration audit re-ran every committed analysis script over the archived captures and reproduced every reported value. The calibration axes regenerated bit-identically. The seeded bootstraps regenerated exactly. The diff over all 23 scripts was clean.

Fixtures. A permanent fixture suite pushes synthetic data with analytically known answers through the actual pipeline functions. Nine of nine fixtures recover their known answers.

Audits. Sign audits confirmed that midpoint-referenced readings have zero own-class sign violations across all 24 layers of both tasks, and that the per-layer accumulated-context axes keep held-out sign with accuracy 0.93–1.00. Boundary audits confirmed contiguous positions 1 to 40 in every run and the correct target token for every carrier. Join audits confirmed that calibration items are drawn from the same pools as the context sentences and that held-out folds separate by construction. Family-level label shuffles kill every effect they are run on (Table 7).

Table 7: Label-shuffle audits. Each effect is recomputed with the class labels shuffled at the scene-family level. The shuffled column gives the permutation band or mean. The minimal-pair row is the fiction/real task; the remnant-gap row is the tank task, averaged over its two directions.

Effect	Unshuffled	Shuffled
Minimal-pair shift (axis units)	+0.99	band [−0.20, +0.21]
Scene-held-out accuracy	0.907	0.530
Remnant gap (axis units)	+1.66	band [−1.42, +1.31]

Capture days. The chat template stamps the capture date into every input (§2.1). The tank transition runs, no-shift runs, and calibration set were captured on 27 August 2026, except two transition runs and one no-shift run captured the next day. Every fiction/real transition run, no-shift run, and calibration item was captured on 28 August. Each task's checkpoint captures fall on one day, as do its mixture-sweep cells and the minimal pairs. Behavior completions were generated on later days, and each completion's reading comes from the same forward pass as the completion. The 2,048-token behavior completions were captured on 5 September 2026 with the template date pinned to each cell's original day, which reproduces the original forward pass exactly; the date-bound run below used 5 September. A committed script reproduces this table from the session manifests.

Generation budget. The behavior completions reported in §3.5 were decoded with a 2,048-token cap and the chat template's date pinned to each cell's original capture day. An earlier pass with a 256-token cap ended inside the model's reasoning channel in 71 of 108 tank and 120 of 204 fiction/real outputs, and its categories had been read from that truncated reasoning; it is superseded. Determinism was checked before the longer pass: the same cell captured twice gives byte-identical text and an identical reading, and every 2,048-token output extends its 256-token text byte for byte with the reading unchanged. Both sets of categories, and the reasoning channel's pre-loop commitment for every cell, are in the repository.

Date-effect bound. The 24 no-shift behavior cells were captured once more with the template date pinned to 5 September 2026 instead of their original day. The reading at the calibrated site moved by at most 0.0024 axis units in the tank task and 0.0182 in the fiction/real task, under one percent of the class separation and an order of magnitude below the smallest effect reported. The greedy completion text diverged in 23 of the 24 cells, from a few characters to a few hundred in, yet where both days delivered an answer the category was the same in all 18 cells. Three fiction/real cells that had looped on their original day delivered an answer on the later one. The date tokens cannot explain any reading effect; they can change whether a particular greedy path loops.

Regeneration instructions and the mapping between the paper's terms and the repository's working names are in the study README.

Appendix C — Supplementary figures

Ten supplementary figures are in the repository's figure directory, alongside a copy of the matched-composition behavior figure of §3.5. They are: the per-run fit gallery for the fiction/real task; the hysteresis loop for the fiction/real sweep; the raw-axis heatmaps for both tasks and the midpoint-referenced heatmap for the fiction/real task, which show the instrument before the corrections of Box 1; the ' word' site just before ' tank' in the carrier; the occupancy bands at layer 4; the jumpiness diagnostic; the norm-against-alignment diagnostic; and the calibration accuracy by layer. The monitor ROC is in the main text.

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